

École Polytechnique Fédérale de Lausanne

Thermal properties of MOFs from molecular dynamics

Master Thesis

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Abstract

Methane-loaded MOF-5 heat capacity is estimated by combining classical NPT molecular dynamics with a loaded-system harmonic quantum correction from PET-JAX/SADMOF Hessians. PET-MAD and PET-SOL calculations at 50, 100, and 150 methane molecules give exploratory room-temperature values of about 0.93, 1.07–1.08, and 1.16 J/g/K. Sampling errors and paired last-layer prediction-rigidity (LLPR) model uncertainties are propagated separately. Production drift, imaginary modes, incomplete relaxation, and sensitivity to uncertainty propagation prevent a claim of convergence. Variation across quenched minima remains unestimated. The empty-framework Hessian provides a separate reference.

1 Scientific motivation and scope

MOF-5 combines high-frequency bond vibrations with soft framework motion; methane adds internal vibrations and anharmonic motion within the pores. Classical MD samples the latter, while normal modes correct the excessive classical occupation of high-frequency vibrations. The conventional host cell contains 424 atoms, $C_{192}H_{96}O_{104}Zn_{32}$; 50, 100, and 150 methane molecules increase the totals to 674, 924, and 1174.

Figure 1 summarizes the calculation. All loaded terms use the same PET checkpoint, composition, and mass normalization. The aim is to test the hybrid approximation of Kapil *et al.* [22] on PET through convergence and validation, rather than matching their force-field results.

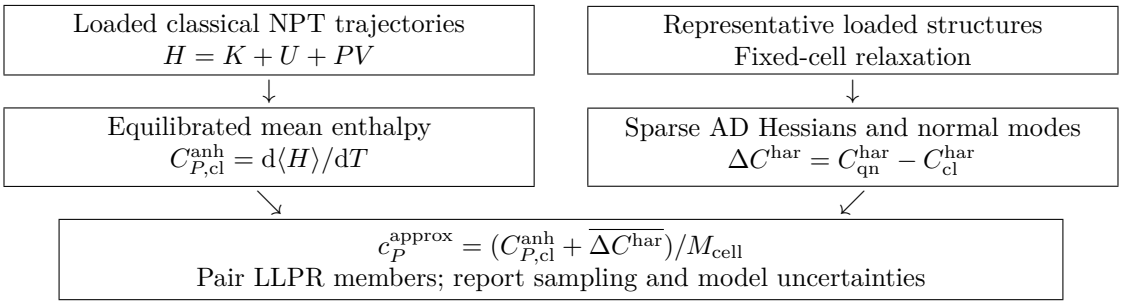


Figure 1: Loaded-system workflow. Accept harmonic references only after force and signed-spectrum checks. The empty-MOF Hessian is evaluated separately, without an empty-MOF MD campaign.

2 Theory

2.1 Quantum Mechanics

2.1.1 The Hamiltonian and the Schrödinger equation

When treating a system of atoms or molecules, we need to treat them as a quantum system. If neglecting relativistic effects, the system is described by a nonrelativistic Hamiltonian, which for electron positions $\boldsymbol{r}$ and nuclear positions $\boldsymbol{R}$ is

$$\hat{\mathcal{H}}_{\mathrm{tot}} = \hat{T}_{\mathrm{n}} + \hat{T}_{\mathrm{e}} + \hat{V}_{\mathrm{ee}} + \hat{V}_{\mathrm{en}} + \hat{V}_{\mathrm{nn}}, \tag{1}$$

$$\hat{T}_{\mathrm{n}} = -\sum_{i=1}^{N} \frac{\hbar^2}{2m_i} \nabla^2_{\boldsymbol{R}_i}, \qquad \hat{T}_{\mathrm{e}} = -\sum_{a=1}^{N_{\mathrm{e}}} \frac{\hbar^2}{2m_{\mathrm{e}}} \nabla^2_{\boldsymbol{r}_a}. \tag{2}$$

The $\hat{T}$ terms represent the kinetic energy of the nuclei and electrons, respectively, while the $\hat{V}$ terms describe electron–electron, electron–nuclear, and nuclear–nuclear Coulomb interactions. To know the state of our quantum system, we need to use the Hamiltonian in the time-dependent Schrödinger equation, which is a partial differential equation for the wavefunction $\Psi(\boldsymbol{r}, \boldsymbol{R}, t)$:

$$i\hbar \frac{\partial \Psi(\boldsymbol{r}, \boldsymbol{R}, t)}{\partial t} = \hat{\mathcal{H}}_{\mathrm{tot}} \Psi(\boldsymbol{r}, \boldsymbol{R}, t). \tag{3}$$

The problem is that solving this equation analytically is only possible for the simplest system, the hydrogen atom. For any other atom or molecule, there is no analytical solution and we have to rely on approximations and numerical methods. All approximations are not created equal, we have to justify using them in each case and we have to think about their feasability. The Born–Oppenheimer approximation [1] is one of the most important approximations in quantum chemistry and molecular physics, and it is the one we will use in this work.

2.1.2 The Born–Oppenheimer approximation

Because nuclei are much heavier than electrons, their motion is much slower than that of the electrons. As the nuclei move along the PES, the electrons adjust instantaneously to the new nuclear positions, which allows us to separate the motion of the nuclei from the motion of the electrons. In other words, we can solve the electronic part of the Schrödinger equation using the electronic Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{el}} = \hat{T}_{\mathrm{e}} + \hat{V}_{\mathrm{ee}} + \hat{V}_{\mathrm{en}} \tag{4}$$

by considering the nuclear positions as fixed parameters in the time-independent Schrödinger equation

$$\hat{\mathcal{H}}_{\mathrm{el}}(\boldsymbol{R})\psi_0(\boldsymbol{r};\boldsymbol{R}) = \varepsilon_0(\boldsymbol{R})\psi_0(\boldsymbol{r};\boldsymbol{R}), \tag{5}$$

$$\tag{6}$$

Solving this equation gives us the electronic ground-state wavefunction $\psi_0(\boldsymbol{r};\boldsymbol{R})$ and the electronic energy $\varepsilon_0(\boldsymbol{R})$, which depends on the nuclear positions parametrically. Using the nuclear Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{nuc}} = -\sum_i \frac{\hbar^2}{2m_i}\nabla^2_{\boldsymbol{R}_i} + U(\boldsymbol{R}), \qquad U(\boldsymbol{R}) = \varepsilon_0(\boldsymbol{R}) + V_{\mathrm{nn}}(\boldsymbol{R}) \tag{7}$$

we can solve the nuclear part of the Schrödinger equation

$$\hat{\mathcal{H}}_{\mathrm{nuc}}\chi(\boldsymbol{R}) = E_n\chi(\boldsymbol{R}), \tag{8}$$

Solving this equation gives us the nuclear wavefunction $\chi(\boldsymbol{R})$ and the nuclear energy E_n. The total wavefunction of the system can then be approximated as the product of the electronic and nuclear wavefunctions:

$$\Psi(\boldsymbol{r},\boldsymbol{R}) \approx \psi_0(\boldsymbol{r};\boldsymbol{R})\chi(\boldsymbol{R}), \tag{9}$$

2.1.3 Classical MD and quantum nuclear statistics

Classical MD further replaces nuclear operators by positions and momenta, giving the Hamiltonian in Eq. (25) and Newtonian forces $m_i\ddot{\boldsymbol{R}}_i = -\nabla_i U$. A useful harmonic criterion and thermal length scale are

$$\beta\hbar\omega \ll 1, \qquad \lambda_i = \sqrt{\frac{2\pi\hbar^2}{m_i k_{\mathrm{B}} T}}, \qquad \beta = (k_{\mathrm{B}}T)^{-1}. \tag{10}$$

Classical sampling is appropriate when quantum level spacing is small relative to thermal energy and nuclear delocalization is negligible on relevant structural scales. A hydrogen-containing system can satisfy these conditions for slow collective modes while violating them for bond vibrations.

Retaining quantum nuclei on the same surface gives the canonical density operator, partition function, and observable average

$$\hat{\rho}_{\mathrm{q}} = Z_{\mathrm{q}}^{-1} e^{-\beta\hat{\mathcal{H}}_{\mathrm{BO}}}, \qquad Z_{\mathrm{q}} = \mathrm{Tr}\, e^{-\beta\hat{\mathcal{H}}_{\mathrm{BO}}} = \sum_n e^{-\beta E_n}, \tag{11}$$

$$\langle A\rangle_{\mathrm{q}} = \mathrm{Tr}(\hat{\rho}_{\mathrm{q}}\hat{A}). \tag{12}$$

The nuclear trace avoids explicit electronic propagation, but not the quantum nuclear problem. PIMD evaluates it through imaginary-time paths: factorizing the exponential and inserting position states produces the ring-polymer distribution in Section 2.4.2.

We will use the Born–Oppenheimer surface for classical MD and also for PIMD to include nuclear quantum effects, especially in H_2O loadings. Hydrogen's small mass increases delocalization; O–H zero-point motion and quantum fluctuations of hydrogen bonds motivate this treatment [8]. Their effect on confined-water heat capacity must be evaluated on the selected potential. PIMD retains anharmonic nuclear sampling, rather than adding a harmonic correction to classical motion. The methane results reported here still use classical MD with the separate harmonic correction; water PIMD is a planned extension.

2.2 Density Functional Theory

At fixed nuclear positions, density functional theory (DFT) expresses the electronic ground-state energy in terms of the electron density $n(\boldsymbol{r})$ rather than the many-electron wavefunction. This provides the Born–Oppenheimer potential-energy surface used to generate reference energies, forces, and stresses for machine-learned potentials.

2.2.1 The Hohenberg–Kohn theorems

For a fixed electron number N_e and external potential $v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})$, the electronic Hamiltonian is

$$\hat{\mathcal{H}}_\mathrm{el} = \hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee} + \sum_{a=1}^{N_\mathrm{e}} v_\mathrm{ext}(\boldsymbol{r}_a;\boldsymbol{R}). \tag{13}$$

The Hohenberg–Kohn (HK) theorems establish the basis of ground-state DFT [6]:

1. The ground-state density $n_0(\boldsymbol{r})$ determines the external potential up to an additive constant, and hence the electronic Hamiltonian and ground-state properties.
2. The ground-state energy is the minimum of the density functional $E_v[n]$, attained at n_0, over admissible densities with $\int n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} = N_\mathrm{e}$.

Writing the universal functional $F[n]$ in constrained-search form makes this variational principle explicit [5]:

$$F[n] = \min_{\psi\to n} \langle\psi|\hat{T}_\mathrm{e} + \hat{V}_\mathrm{ee}|\psi\rangle, \tag{14}$$

$$E_v[n] = F[n] + \int v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}, \qquad E_0 = \min_{n \text{ admissible}} E_v[n]. \tag{15}$$

HK therefore turns the electronic problem into a density minimization, but does not supply a practical expression for $F[n]$.

2.2.2 From HK to the Kohn–Sham equations

Kohn–Sham (KS) theory introduces an auxiliary system of noninteracting electrons with the same density [7]. Its orthonormal orbitals ϕ_a and occupations f_a give

$$n(\boldsymbol{r}) = \sum_a f_a|\phi_a(\boldsymbol{r})|^2, \qquad \sum_a f_a = N_\mathrm{e}. \tag{16}$$

The occupations include spin multiplicity; for a periodic solid the sum also includes band indices and wavevector weights. The unknown HK functional is decomposed as

$$F[n] = T_\mathrm{s}[n] + E_\mathrm{H}[n] + E_\mathrm{xc}[n], \tag{17}$$

$$T_\mathrm{s}[n] = \sum_a f_a \int \phi_a^*(\boldsymbol{r}) \left(-\frac{\hbar^2}{2m_\mathrm{e}}\nabla^2\right) \phi_a(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}, \tag{18}$$

$$E_\mathrm{H}[n] = \frac{1}{2}\iint n(\boldsymbol{r})w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{r}'. \tag{19}$$

Here T_s is the noninteracting kinetic energy and E_H is the classical electron–electron electrostatic energy. For the periodic MOF, w is the periodic Coulomb kernel. The exchange–correlation functional E_xc contains the remaining interaction energy and the correction to T_s. Thus the electronic energy functional becomes

$$E_\mathrm{el}[n;\boldsymbol{R}] = T_\mathrm{s}[n] + \int v_\mathrm{ext}(\boldsymbol{r};\boldsymbol{R})n(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} + E_\mathrm{H}[n] + E_\mathrm{xc}[n]. \tag{20}$$

Minimizing this energy with respect to the orbitals while preserving their orthonormality gives the one-electron KS equations:

$$\left[-\frac{\hbar^2}{2m_\mathrm{e}}\nabla^2 + v_\mathrm{eff}[n](\boldsymbol{r})\right]\phi_a(\boldsymbol{r}) = \epsilon_a\phi_a(\boldsymbol{r}), \tag{21}$$

$$v_\mathrm{eff}[n](\boldsymbol{r}) = v_\mathrm{ext}(\boldsymbol{r}) + v_\mathrm{H}[n](\boldsymbol{r}) + v_\mathrm{xc}[n](\boldsymbol{r}), \tag{22}$$

$$v_\mathrm{H}[n](\boldsymbol{r}) = \int w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}', \qquad v_\mathrm{xc}[n](\boldsymbol{r}) = \frac{\delta E_\mathrm{xc}}{\delta n(\boldsymbol{r})}. \tag{23}$$

The decomposition is formally exact when the auxiliary system can reproduce the density; practical DFT approximates E_xc. Because v_eff depends on n, the equations are solved self-consistently: start from a trial density, solve for the orbitals, rebuild the density, and iterate until both density and energy converge.

At the converged density n_0, adding nuclear repulsion gives the potential energy and its derivatives give the reference forces:

$$U_\mathrm{DFT}(\boldsymbol{R}) = E_\mathrm{el}[n_0;\boldsymbol{R}] + V_\mathrm{nn}(\boldsymbol{R}), \qquad \boldsymbol{F}_i^\mathrm{DFT} = -\nabla_{\boldsymbol{R}_i} U_\mathrm{DFT}|_{\boldsymbol{h}}\,. \tag{24}$$

Cell-strain derivatives similarly provide stresses. These quantities supply training labels for MLIPs such as PET-MAD [24]. In this project, the MLIP then evaluates the potential-energy surface during MD and Hessian calculations without repeating the electronic calculation.

2.3 Statistical Mechanics

The classical simulations in this project use the isothermal–isobaric (NPT) ensemble. The numbers of framework atoms and guest molecules are fixed, while the system exchanges energy and volume with reservoirs at temperature T and imposed hydrostatic pressure P. Allowing the periodic cell to fluctuate captures the volume response of the loaded framework. The quantity of interest is therefore the constant-pressure heat capacity at fixed loading [2, 3].

For N classical atoms, write the positions and momenta explicitly as $\boldsymbol{r}^N = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_N)$ and $\boldsymbol{p}^N = (\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N)$. The cell dependence of functions is left implicit throughout this section: they are evaluated in the instantaneous periodic cell, with $V = \det \boldsymbol{h}$. The cell remains an independent fluctuating variable; positions and momenta alone do not determine its volume. We denote the physical Hamiltonian by E, reserving H for enthalpy:

$$E(\boldsymbol{r}^N, \boldsymbol{p}^N) = K(\boldsymbol{p}^N) + U(\boldsymbol{r}^N) = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{r}^N), \tag{25}$$

where U is the MLIP potential energy. The instantaneous physical enthalpy is

$$H(\boldsymbol{r}^N, \boldsymbol{p}^N) = E(\boldsymbol{r}^N, \boldsymbol{p}^N) + PV = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m_i} + U(\boldsymbol{r}^N) + PV. \tag{26}$$

It uses the imposed pressure and excludes thermostat and barostat energies.

With $\beta = (k_\mathrm{B}T)^{-1}$, the NPT partition function and probability density are

$$\Delta(N, P, T) = \mathcal{C} \int_0^\infty \frac{\mathrm{d}V}{V_0} \int_{\mathrm{cell}(V)} \mathrm{d}^{3N}r \int \mathrm{d}^{3N}p \, e^{-\beta H(\boldsymbol{r}^N, \boldsymbol{p}^N)}, \tag{27}$$

$$\rho(\boldsymbol{r}^N, \boldsymbol{p}^N) = \Delta^{-1} e^{-\beta H(\boldsymbol{r}^N, \boldsymbol{p}^N)}. \tag{28}$$

Here V_0 is a fixed, temperature- and pressure-independent reference volume, and $\mathcal{C}$ contains the fixed phase-space normalization factors needed to make Δ dimensionless. The position integral covers the instantaneous cell for every atom, and the momentum integral covers the allowed momenta. The volume integral is retained because the cell fluctuates in NPT. Although the cell argument is suppressed in ρ, this is the joint density of atomic positions, momenta, and cell volume, rather than a density with the volume already integrated out. Every equilibrium observable is averaged as

$$\langle A \rangle = \mathcal{C} \int_0^\infty \frac{\mathrm{d}V}{V_0} \int_{\mathrm{cell}(V)} \mathrm{d}^{3N}r \int \mathrm{d}^{3N}p \, \rho(\boldsymbol{r}^N, \boldsymbol{p}^N) A(\boldsymbol{r}^N, \boldsymbol{p}^N), \tag{29}$$

Throughout this report, P denotes the imposed hydrostatic pressure, and equilibrium averages, variances, and covariances refer to the NPT ensemble unless stated otherwise. Ensemble subscripts are omitted.

The corresponding thermodynamic potential is the Gibbs free energy. For a temperature-independent Hamiltonian,

$$G = -k_\mathrm{B}T \ln \Delta, \qquad \langle H \rangle = -\left(\frac{\partial \ln \Delta}{\partial \beta} \right)_{N,P}, \tag{30}$$

$$\langle V \rangle = \left(\frac{\partial G}{\partial P} \right)_{N,T}, \qquad S = -\left(\frac{\partial G}{\partial T} \right)_{N,P}, \tag{31}$$

$$\langle E \rangle = \langle H \rangle - P\langle V \rangle, \qquad G = \langle H \rangle - TS. \tag{32}$$

The volume integral above displays the scalar-volume formulation. In the actual simulations the cell matrix $\boldsymbol{h}$ also permits anisotropic strain under hydrostatic external pressure, with $V = \det \boldsymbol{h}$. Fractional coordinates $\boldsymbol{s}_i = \boldsymbol{h}^{-1}\boldsymbol{r}_i$ give $\mathrm{d}^{3N}r = V^N \mathrm{d}^{3N}s$; integrating out the momenta therefore gives a configurational weight proportional to $V^N e^{-\beta[U+PV]}$ in fractional coordinates. Flexible-cell sampling must include the appropriate cell measure and extended-system corrections [27].

Additionally, we assume the ergodic hypothesis, which states that the infinite time average of an observable along a trajectory is equal to its ensemble average. So the integral in Eq. (29) can be replaced by a time average along a sufficiently long trajectory:

$$\langle A \rangle = \lim_{\tau \to \infty} \frac{1}{\tau} \int_0^\tau A\big(\boldsymbol{r}^N(t), \boldsymbol{p}^N(t)\big) \, \mathrm{d}t. \tag{33}$$

In practice, we can't run the simulation for ever, so we need to run the simulation for long enough to ensure that the system has reached equilibrium and explored the relevant phase space. We can then replace the integral by a finite sum over the trajectory frames, giving the time average

$$\langle A \rangle \approx \frac{1}{M} \sum_{j=1}^{M} A\big(\boldsymbol{r}^N(t_j), \boldsymbol{p}^N(t_j)\big), \tag{34}$$

2.3.1 Equipartition and temperature

Each independent quadratic term contributes $k_{\mathrm{B}}T/2$. For f active momentum degrees of freedom,

$$\langle K\rangle = \frac{f}{2}k_{\mathrm{B}}T, \qquad T_{\mathrm{kin}} = \frac{2K}{fk_{\mathrm{B}}}. \tag{35}$$

Subtract constraints and deliberately removed momentum components from $3N$; fixing total linear momentum gives $f = 3N - 3$. Auxiliary reservoir variables do not count. Initial velocities follow the Maxwell–Boltzmann distribution:

$$v_{i\alpha} \sim \mathcal{N}\left(0, \frac{k_{\mathrm{B}}T}{m_i}\right), \qquad \alpha \in \{x, y, z\}. \tag{36}$$

Subtract center-of-mass velocity when required. The NPT momentum distribution gives $\mathrm{var}(K) = f(k_{\mathrm{B}}T)^2/2$ and $\mathrm{var}(T_{\mathrm{kin}}) = 2T^2/f$; repeated exact rescaling suppresses these fluctuations.

2.3.2 Response functions and heat capacity

For an observable A with no explicit temperature dependence and a temperature-independent Hamiltonian, differentiating the NPT average gives

$$\left(\frac{\partial\langle A\rangle}{\partial T}\right)_{N,P} = \mathrm{cov}\left(A, \frac{\partial \ln\rho}{\partial T}\right) = \frac{\mathrm{cov}(A,H)}{k_{\mathrm{B}}T^2}, \tag{37}$$

Taking $A = H$ gives the constant-pressure heat capacity:

$$C_P = \left(\frac{\partial\langle H\rangle}{\partial T}\right)_{N,P} = \frac{\mathrm{var}(H)}{k_{\mathrm{B}}T^2}. \tag{38}$$

For separable kinetic energy, the momenta are independent of the configuration and volume. The equivalent configurational estimator is

$$C_P = \frac{f}{2}k_{\mathrm{B}} + \frac{\mathrm{var}(U+PV)}{k_{\mathrm{B}}T^2}. \tag{39}$$

The kinetic contribution $fk_{\mathrm{B}}/2$ must be included when using potential-energy and volume fluctuations alone.

The volume responses are also expressed entirely through NPT averages:

$$\kappa_T = -\frac{1}{\langle V\rangle}\left(\frac{\partial\langle V\rangle}{\partial P}\right)_{N,T} = \frac{\mathrm{var}(V)}{k_{\mathrm{B}}T\langle V\rangle}, \tag{40}$$

$$\alpha_V = \frac{1}{\langle V\rangle}\left(\frac{\partial\langle V\rangle}{\partial T}\right)_{N,P} = \frac{\mathrm{cov}(V,H)}{k_{\mathrm{B}}T^2\langle V\rangle}. \tag{41}$$

Explicitly state-dependent effective potentials require additional response terms. The present workflow instead differentiates mean NPT enthalpies across temperatures (Eq. (120)). Smaller temperature spacing reduces discretization bias but amplifies noise. Cell heat capacities become mass-specific values through $c_P = C_P/M_{\mathrm{cell}}$, including guest mass.

2.3.3 Quantum harmonic heat capacity and the Dulong–Petit law

Quantum statistical mechanics replaces phase-space integrals by traces over nuclear quantum states. The quantum counterpart of the pressure ensemble is

$$\Delta_{\mathrm{q}} = \int_0^\infty \frac{\mathrm{d}V}{V_0}\ \mathrm{Tr}\, e^{-\beta[\hat{E}(\hat{\boldsymbol{r}}^N,\hat{\boldsymbol{p}}^N)+PV]}, \qquad \hat{\rho} = \Delta_{\mathrm{q}}^{-1} e^{-\beta[\hat{E}(\hat{\boldsymbol{r}}^N,\hat{\boldsymbol{p}}^N)+PV]}. \tag{42}$$

An average is obtained by integrating $\mathrm{Tr}[\hat{\rho}\hat{A}]$ over $\mathrm{d}V/V_0$, with the cell argument again implicit. The enthalpy derivative therefore defines C_P as before. In this project, however, the quantum correction is evaluated at a fixed optimized cell, where the harmonic modes have fixed frequencies. The following averages are explicitly for this fixed-cell quantum harmonic reference, and give C_V; the classical MD contribution remains at constant pressure.

Expanding the potential about a stable minimum and using mass-weighted normal coordinates Q_k gives independent harmonic oscillators:

$$\hat{E}_{\mathrm{har}} = U_{\mathrm{min}} + \sum_{k\in\mathcal{V}}\left(\frac{\hat{\Pi}_k^2}{2} + \frac{\omega_k^2\hat{Q}_k^2}{2}\right), \qquad \epsilon_{k,n} = \hbar\omega_k\left(n+\frac{1}{2}\right), \quad n = 0, 1, \ldots. \tag{43}$$

Here $\mathcal{V}$ contains the retained positive-frequency modes and Π_k is conjugate to Q_k. Each mode has partition function

$$z_k = \sum_{n=0}^{\infty} e^{-\beta\epsilon_{k,n}} = \frac{e^{-x_k/2}}{1-e^{-x_k}}, \qquad x_k = \beta\hbar\omega_k, \qquad Z_{\mathrm{har}} = e^{-\beta U_{\min}} \prod_{k\in\mathcal{V}} z_k. \tag{44}$$

Differentiating its logarithm gives the mean energy:

$$\langle E\rangle_{\mathrm{har}} = -\frac{\partial \ln Z_{\mathrm{har}}}{\partial\beta} = U_{\min} + \sum_{k\in\mathcal{V}} \hbar\omega_k \left(\frac{1}{2} + \frac{1}{e^{x_k}-1}\right). \tag{45}$$

The zero-point energy $\hbar\omega_k/2$ is independent of temperature at fixed frequencies and does not contribute to the heat capacity. Differentiating the thermal occupation gives

$$C_{V,\mathrm{q}}^{\mathrm{har}} = \left(\frac{\partial\langle E\rangle_{\mathrm{har}}}{\partial T}\right)_V = \sum_{k\in\mathcal{V}} k_{\mathrm{B}} \frac{x_k^2 e^{x_k}}{(e^{x_k}-1)^2}. \tag{46}$$

Modes with $\hbar\omega_k \gg k_{\mathrm{B}}T$ are thermally suppressed, whereas low-frequency modes approach their classical contribution.

In the high-temperature limit, $x_k \ll 1$, using $e^{x_k} - 1 \simeq x_k$ gives $C_{V,k} \to k_{\mathrm{B}}$. Classical equipartition gives the same result: each oscillator has one quadratic kinetic and one quadratic potential term, together contributing $k_{\mathrm{B}}T$ to its mean energy. Thus

$$C_{V,\mathrm{cl}}^{\mathrm{har}} = |\mathcal{V}|k_{\mathrm{B}}, \qquad C_V \longrightarrow 3Nk_{\mathrm{B}}, \qquad C_{V,\mathrm{molar}} \longrightarrow 3R, \quad R = N_{\mathrm{A}}k_{\mathrm{B}}. \tag{47}$$

The last two expressions are the Dulong–Petit law for a bulk solid with N atoms, with molar heat capacity defined per mole of atoms. A finite periodic-cell calculation omitting the three uniform translations instead has $3N-3$ vibrational modes before any additional mode filtering. For the loaded MOF, N includes both framework and guest atoms; the mass-specific limit is $|\mathcal{V}|k_{\mathrm{B}}/M_{\mathrm{cell}}$.

The project uses these results to replace the classical harmonic contribution by its quantum value:

$$C_P^{\mathrm{hyb}}(T) \simeq C_{P,\mathrm{cl}}^{\mathrm{anh}}(T) + \sum_{k\in\mathcal{V}} k_{\mathrm{B}} \left[\frac{x_k^2 e^{x_k}}{(e^{x_k}-1)^2} - 1\right]. \tag{48}$$

The correction vanishes in the classical high-temperature limit. Adding this fixed-cell C_V correction to the MD C_P assumes that the quantum change in $C_P - C_V$ is small.

2.3.4 Uncertainty Quantification

Discard initial transients using running/block averages, structure, and replica comparisons. Minimization does not replace thermal equilibration. For M equilibrated NPT production samples saved every δt_{s},

$$\overline{A} = \frac{1}{M}\sum_{n=1}^{M} A_n, \tag{49}$$

$$C_A(k) = \langle (A_n - \langle A\rangle)(A_{n+k} - \langle A\rangle)\rangle, \tag{50}$$

$$\rho_A(k) = C_A(k)/C_A(0). \tag{51}$$

The sample-mean variance is

$$\mathrm{var}(\overline{A}) = \frac{C_A(0)}{M}\left[1 + 2\sum_{k=1}^{M-1}\left(1 - \frac{k}{M}\right)\rho_A(k)\right]. \tag{52}$$

For runs much longer than the correlation time,

$$g_A = 1 + 2\sum_{k=1}^{\infty}\rho_A(k), \qquad \tau_{\mathrm{int},A} = \frac{\delta t_{\mathrm{s}}}{2} g_A, \tag{53}$$

$$M_{\mathrm{eff},A} \simeq \frac{M}{g_A}, \qquad \mathrm{SE}(\overline{A}) \simeq \sqrt{\frac{C_A(0)}{M_{\mathrm{eff},A}}}. \tag{54}$$

Estimate or truncate the correlation sum statistically. Effective sample counts depend on the observable; finer output stride does not add independent information over a fixed duration.

Block errors should plateau for blocks longer than the correlation time, while retaining enough blocks [12]. Independent replicas help detect slow modes, but neither method reveals a basin never visited. Resample whole replicas or sufficiently long blocks and recompute the full estimator, preserving cross-temperature correlations. Variance-based heat capacities require errors of squared fluctuations, not merely errors of the mean enthalpy. Sampling error, finite-size effects, integration bias, and potential error need separate checks.

2.3.5 Configurational populations

For temperature-independent basins $\mathcal{B}_a$ of configuration and cell space, their restricted NPT partition functions are

$$\Delta_a = \mathcal{C} \int_{\mathcal{B}_a} \frac{\mathrm{d}V}{V_0}\, \mathrm{d}^{3N} r\, \mathrm{d}^{3N} p\, e^{-\beta H(\boldsymbol{r}^N, \boldsymbol{p}^N)}. \tag{55}$$

$$\Delta = \sum_a \Delta_a, \qquad w_a = \frac{\Delta_a}{\Delta}, \qquad \langle H \rangle = \sum_a w_a \langle H \rangle_a. \tag{56}$$

$$C_P = \sum_a w_a \left(\frac{\partial \langle H \rangle_a}{\partial T} \right)_{N,P} + \sum_a \langle H \rangle_a \left(\frac{\partial w_a}{\partial T} \right)_{N,P}. \tag{57}$$

Here $\langle H \rangle_a$ is the conditional NPT mean in basin a. The two terms describe motion within basins and redistribution between them. A Hessian characterizes one local environment; adequately sampled MD also captures changing guest arrangements and framework conformations. Basin weights depend on accessible phase-space volume, so counting optimized structures does not establish equilibrium populations.

2.4 Molecular Dynamics

2.4.1 Classical molecular dynamics

Classical molecular dynamics (MD) generates a trajectory by integrating the nuclear equations of motion on the MLIP potential-energy surface. The framework and guests move together, so the trajectory samples bond vibrations, framework deformation, guest motion, and changes of adsorption environment. Unlike a harmonic expansion, MD evaluates the full potential at each configuration and can therefore capture classical anharmonicity. The role of MD here is to estimate equilibrium enthalpies at fixed loading and pressure, rather than to solve the electronic problem at each step [2, 3].

1. Specify the physical system and forces. For atom i with mass m_i, position $\boldsymbol{R}_i$, and momentum $\boldsymbol{p}_i$, the fixed-cell physical equations are

$$\dot{\boldsymbol{R}}_i = \frac{\boldsymbol{p}_i}{m_i}, \qquad \dot{\boldsymbol{p}}_i = \boldsymbol{F}_i, \qquad \boldsymbol{F}_i = -\nabla_{\boldsymbol{R}_i} U(\boldsymbol{R}; \boldsymbol{h})\big|_{\boldsymbol{h}}. \tag{58}$$

The masses set the inertial response; the model supplies the potential energy, forces, and cell derivatives. The physical energy is $E = K + U$, where $K = \sum_i |\boldsymbol{p}_i|^2/(2m_i)$. Without reservoir coupling, an exact fixed-cell trajectory conserves E. The production calculation adds thermostat and barostat variables to sample the pressure ensemble defined in Section 2.3.

The chosen host structure, guest species, loading, model, and atom masses define a state point. Guest molecules remain in the periodic system: there is no exchange of particles with a gas reservoir. All framework and guest atoms are flexible; molecular rotations and internal vibrations emerge from atomic motion, without separate rigid-body equations. The model must cover these configurations and provide mutually consistent energies, forces, and stresses. A model with energies and forces alone is insufficient for the flexible-cell calculation.

2. Represent the periodic cell and atomic neighborhoods. The columns of $\boldsymbol{h}$ are the cell vectors, $V = \det \boldsymbol{h}$, and fractional coordinates obey $\boldsymbol{R}_i = \boldsymbol{h}\boldsymbol{s}_i$. Periodic images have positions $\boldsymbol{R}_i + \boldsymbol{h}\boldsymbol{n}$, $\boldsymbol{n} \in \mathbb{Z}^3$. A periodic pair displacement is

$$\boldsymbol{r}_{ij\boldsymbol{n}} = \boldsymbol{R}_j - \boldsymbol{R}_i + \boldsymbol{h}\boldsymbol{n}. \tag{59}$$

Wrapping fractional coordinates into a unit cell changes their displayed image, not the physical configuration. Unwrapped coordinates retain image history and are needed when following guest displacements across boundaries.

For a single-image cutoff, r_c must be smaller than half the shortest nonzero lattice-vector length; otherwise all relevant images within the cutoff must be included. This matters for the short or skew cells of some supplied MOFs. In a triclinic cell, rounding fractional displacements alone need not find the geometrically nearest image. The neighborhood construction must respect the actual cell and periodic image labels throughout its deformation.

A neighbor list stores candidate interactions out to $r_c + r_{\mathrm{skin}}$. The skin avoids rebuilding the list after every small displacement, while rebuild checks ensure that new interactions are found before atoms cross into the physical cutoff. Both atomic motion and cell deformation can alter neighbors. The current LAMMPS writer uses a 2 Å skin and checks rebuilding every step. Cutoffs and model derivatives must remain smooth; changing the neighbor list must not introduce artificial force jumps. Any additional long-range term would need its own consistent periodic treatment.

3. Relax the inserted structure before assigning velocities. Random guest insertion can leave high-energy contacts even after geometric separation checks. Fresh loaded-system runs first minimize U at fixed cell using FIRE. This damped optimization uses auxiliary velocities and an adaptive step to approach a local minimum; it is not a finite-temperature trajectory [11]. The initial cell is held fixed, so this stage relaxes atomic positions without yet determining the equilibrium volume at the target pressure.

The input records the force tolerance, displacement bound, and iteration and force-evaluation budgets. With the configured infinity-norm criterion, the acceptance check is

$$F_{\max} = \max_{i,\alpha} |F_{i\alpha}| \le F_{\mathrm{tol}}. \tag{60}$$

Reaching the iteration budget alone is not convergence. Failure to meet the force criterion, loss of atoms, or nonfinite quantities must stop the run before thermal sampling. After FIRE, the code resets the step counter and restores the specified MD timestep, since FIRE adapts its own step. This initialization is distinct from the later fixed-cell quench of an equilibrated thermal frame used to construct the harmonic reference.

4. Initialize the thermal momenta. After relaxation, fresh runs draw Gaussian atomic velocities at the target T, as specified by Eq. (35) and the Maxwell variance $k_{\mathrm{B}}T/m_i$. The LAMMPS initialization removes net linear and angular momentum. The seed is recorded so the initialization can be reproduced; independent replicas should use independent placements and velocity seeds. The temperature estimator must use the active degrees of freedom, excluding any deliberately removed components or constraints. Initial temperature assignment does not equilibrate the positions, volume, or guest populations: those relax during the subsequent trajectory.

5. Integrate forces with a finite timestep. The basic fixed-cell velocity-Verlet update is

$$\boldsymbol{p}_i^{n+1/2} = \boldsymbol{p}_i^{n} + \frac{\Delta t}{2}\boldsymbol{F}_i(\boldsymbol{R}^n), \tag{61}$$

$$\boldsymbol{R}_i^{n+1} = \boldsymbol{R}_i^{n} + \Delta t\,\frac{\boldsymbol{p}_i^{n+1/2}}{m_i}, \tag{62}$$

$$\boldsymbol{p}_i^{n+1} = \boldsymbol{p}_i^{n+1/2} + \frac{\Delta t}{2}\boldsymbol{F}_i(\boldsymbol{R}^{n+1}). \tag{63}$$

The first half step advances the momenta using the existing force; the position update gives the next configuration; a new model evaluation gives the force for the final half step. This second-order, reversible, symplectic scheme requires one new force evaluation per step once the initial force is known. The production integrator extends this splitting with thermostat and cell updates; the fixed-cell formulas above explain the underlying force–motion cycle, not the complete barostatted algorithm.

The timestep must resolve the fastest motion, particularly flexible C–H and O–H stretches. For an isolated harmonic mode, the stability condition is $\omega_{\max}\Delta t < 2$, but stability alone does not ensure accurate enthalpies. The template uses $\Delta t = 0.5$ fs; its adequacy requires convergence checks. Short fixed-cell energy-conservation tests diagnose integration bias, and comparison of enthalpies and structural observables at smaller steps tests its effect on the final property. A thermostat can remove numerical heating and conceal an inaccurate integrator. Adding bond constraints would change the sampled degrees of freedom and require a matching change to the classical and harmonic heat-capacity terms.

6. Couple the atoms and cell to temperature and pressure. The active production path uses Nosé–Hoover chains rather than stochastic noise. For illustration, a single thermostat variable ξ modifies fixed-cell momenta according to

$$\dot{\boldsymbol{p}}_i = \boldsymbol{F}_i - \xi\boldsymbol{p}_i, \qquad \dot{\xi} = \frac{2K - fk_{\mathrm{B}}T}{Q}, \qquad Q \sim fk_{\mathrm{B}}T\tau_T^2. \tag{64}$$

When K is too large, the thermostat tends to increase friction; when it is too small, it tends to restore kinetic energy. A chain thermostats the auxiliary variables as well, improving sampling of nearly harmonic systems [28]. These variables control sampling but do not contribute to the physical enthalpy. Temperature must fluctuate around its target; forcing its instantaneous value to be constant would suppress equilibrium fluctuations.

Pressure control requires the kinetic contribution and the configurational virial. For unconstrained atoms with bulk streaming removed,

$$\boldsymbol{P}_{\mathrm{inst}} = \frac{1}{V}\left[\sum_i \frac{\boldsymbol{p}_i \otimes \boldsymbol{p}_i}{m_i} + \boldsymbol{W}\right], \qquad P_{\mathrm{inst}} = \frac{1}{3}\,\mathrm{tr}\,\boldsymbol{P}_{\mathrm{inst}}. \tag{65}$$

Here $\otimes$ denotes an outer product. Pair forces give $\boldsymbol{W} = \sum_{i<j} \boldsymbol{r}_{ij} \otimes \boldsymbol{F}_{ij}$; the many-body MLIP instead provides the virial through strain derivatives at fixed fractional positions:

$$W_{\alpha\beta} = -\left.\frac{\partial U}{\partial \epsilon_{\alpha\beta}}\right|_{\boldsymbol{\epsilon}=0,\{\boldsymbol{s}_i\}}, \qquad \boldsymbol{h} \mapsto (\boldsymbol{I} + \boldsymbol{\epsilon})\boldsymbol{h}. \tag{66}$$

Tensile-positive stress has the opposite sign to pressure. The model's configurational stress must be combined with the kinetic term to obtain the full pressure [9].

The flexible-cell barostat responds to the difference between instantaneous and target stress. Hydrostatic loading imposes $P\boldsymbol{I}$, with zero target shear, while allowing the three lengths and three tilt factors to vary independently. Equal normal targets therefore do not force equal strain along each axis. At fixed fractional coordinates, a cell change carries atoms with it through $\boldsymbol{R}_i = \boldsymbol{h}\boldsymbol{s}_i$; additional atomic motion changes $\boldsymbol{s}_i$. The periodic geometry, neighbors, and forces must be updated consistently as both evolve.

LAMMPS implements this through `fix npt` with `tri`, thermostat chains on both atoms and barostat, and `mtk yes`. The latter supplies the measure correction required for the intended pressure ensemble [27, 10]. The template uses a target of 1 bar, chain length three, and temperature/pressure damping times of 100/1000 fs. These are coupling times, not the integration timestep. Fast coupling can excite artificial oscillations; slow coupling can delay equilibration. Energy E is not conserved during reservoir exchange, and fluctuations in temperature and instantaneous pressure are expected. Acceptance rests on their stationary distributions and time averages, together with structural stability.

7. Equilibrate, collect data, and continue interrupted runs. The freshly initialized system evolves under the same temperature and pressure targets during equilibration and production. Equilibration permits energy redistribution, volume adjustment, and guest rearrangement. Discarding a fixed interval is a starting prescription: running averages and block averages of enthalpy and cell parameters, framework geometry, guest motion, and comparisons across replicas must establish stationarity. A settled kinetic temperature alone does not demonstrate equilibration of slow framework or adsorption modes.

The template requests 10^6 steps, or 500 ps, with the first 2×10^5 steps (100 ps) assigned to equilibration. The trajectory dump begins afterward, while the thermodynamic log also includes the early interval. Saving every 100 steps gives 0.05 ps spacing, but does not make successive frames independent. The log contains U, K, temperature, pressure components, volume, density, and cell dimensions; trajectory frames contain atom identities, positions, velocities, forces, and cell geometry. These complementary outputs must be matched by timestep.

Restart files preserve the dynamical state needed for continuation. A continuation reads the latest numeric restart, skips initial relaxation and velocity assignment, appends outputs, and advances toward the original absolute final-step target. It is the same trajectory, not an independent replica. Wall-time continuations extend sampling without resetting the scientific initialization; physical or numerical failures require diagnosis.

8. Obtain the classical heat capacity and harmonic references. For each retained production sample, form the physical enthalpy

$$H_n = K_n + U_n + PV_n, \qquad \overline{H}(T) = \frac{1}{M}\sum_{n=1}^{M} H_n. \tag{67}$$

Use the imposed P, not the fluctuating P_{inst}, and convert all terms to consistent energy units. Neither thermostat nor barostat energies enter H_n. The mean contains both framework and guest contributions, with fixed composition at every temperature. Repeat the calculation over the temperature grid and estimate

$$C_{P,\mathrm{cl}}^{\mathrm{anh}}(T) \simeq \frac{\overline{H}(T+\delta T) - \overline{H}(T-\delta T)}{2\delta T}, \qquad c_{P,\mathrm{cl}}^{\mathrm{anh}} = \frac{C_{P,\mathrm{cl}}^{\mathrm{anh}}}{M_{\mathrm{cell}}}. \tag{68}$$

The default $\delta T = 25$ K grid runs from 175 to 425 K, providing centered derivatives over the reported 200–400 K range. Differentiate replica-averaged enthalpies when replicas are available. Sampling errors must account for autocorrelation and be propagated through the derivative (Section 2.3); finer temperature spacing reduces finite-difference bias but increases sensitivity to noisy means.

Finally, select representative equilibrated loaded structures for the separate fixed-cell minimization and Hessian stage. The shared source is 400 K for the default reporting grid. Its modes supply the harmonic quantum-minus-classical correction described in Section 2.3.3. The empty framework is relaxed directly for its separate reference; it receives no MD campaign. This division assigns classical anharmonic sampling to MD and quantum harmonic populations to the Hessian calculation.

2.4.2 Path integral molecular dynamics

PIMD samples the quantum nuclear statistics of Section **??** on the full anharmonic Born–Oppenheimer surface. We will use it to assess quantum effects in water-loaded MOFs, particularly those associated with hydrogen. Present methane results use classical MD plus a harmonic correction. The following assumes distinguishable nuclei and neglects exchange permutations [3].

Table 1: Classical MD stages and their acceptance checks. Reported campaign settings appear in Section 4.3.

Stage	Required check
Define and insert	Fixed composition, exact periodic geometry, separation, correct species/masses, and model force/stress capability.
Initial relaxation	Fixed cell; accepted force convergence before thermal velocities are assigned.
Integrate and couple	Resolved fast vibrations, consistent periodic neighbors, and validated timestep and reservoir coupling times.
Equilibrate	Stationary enthalpy, cell, framework, and guest observables; sufficient exploration of relevant configurations.
Collect and continue	Physical observables, consistent timesteps, restarts, and preserved continuation history.
Analyze and correct	Correlated sampling errors, temperature derivatives, loaded-mass normalization, and stable harmonic reference minima.

Ring-polymer mapping Starting from Eq. (11) at fixed cell, write $\hat{\mathcal{H}} = \hat{\mathcal{H}}_{\mathrm{BO}} = \hat{K} + \hat{U}$. The noncommuting nuclear kinetic and potential operators are split into L imaginary-time slices, $\beta_L = \beta/L$:

$$e^{-\beta\hat{\mathcal{H}}} = \lim_{L\to\infty}\left[e^{-\beta_L\hat{U}/2}e^{-\beta_L\hat{K}}e^{-\beta_L\hat{U}/2}\right]^L. \tag{69}$$

Inserting position states gives

$$Z_L = \mathcal{A}_L\int\prod_{j=1}^{L}\mathrm{d}^{3N}R^{(j)}\,e^{-\beta U_L^{\mathrm{eff}}}, \tag{70}$$

$$\mathcal{A}_L = \prod_{i=1}^{N}\left(\frac{m_iL}{2\pi\beta\hbar^2}\right)^{3L/2}, \tag{71}$$

$$\begin{aligned} U_L^{\mathrm{eff}} &= \frac{1}{L}\sum_{j=1}^{L}U(\boldsymbol{R}^{(j)};\boldsymbol{h}) \\ &\quad + \frac{1}{L}\sum_{i,j}\frac{m_i\omega_L^2}{2}|\boldsymbol{R}_i^{(j)} - \boldsymbol{R}_i^{(j+1)}|^2, \end{aligned} \tag{72}$$

$$\omega_L = \frac{L}{\beta\hbar}, \qquad \boldsymbol{R}_i^{(L+1)} = \boldsymbol{R}_i^{(1)}. \tag{73}$$

Each nucleus becomes a closed ring of L beads connected by springs; the physical potential acts within each slice. Beads resolve imaginary time, not a real trajectory. Periodic paths need consistent images and, where relevant, winding sectors. The equivalent convention samples LU_L^{eff} at LT. At $L = 1$ the classical configuration weight is recovered; the symmetric primitive error is $\mathcal{O}(L^{-2})$ for smooth potentials [3, 13].

Sampling and observables Auxiliary momenta sample

$$\mathcal{H}_{\mathrm{samp}} = \sum_{i,j}\frac{|\boldsymbol{\pi}_i^{(j)}|^2}{2\mu_i^{(j)}} + U_L^{\mathrm{eff}}. \tag{74}$$

Fictitious masses μ affect efficiency; physical masses m in the springs determine quantum delocalization. Normal-mode thermostats such as PILE handle stiff springs [13]. PIMD simulation time is only a sampling variable; quantum transport or rates require additional real-time approximations such as RPMD/CMD.

For position-diagonal observables,

$$\langle A\rangle_{\mathrm{q}} = \lim_{L\to\infty}\left\langle\frac{1}{L}\sum_{j=1}^{L}A(\boldsymbol{R}^{(j)})\right\rangle_L, \tag{75}$$

A nonlinear observable at the centroid generally differs from the bead average. A centroid-virial physical energy estimator is

$$\overline{\boldsymbol{R}}_i = \frac{1}{L}\sum_j\boldsymbol{R}_i^{(j)}, \tag{76}$$

$$K_{\mathrm{cv}} = \frac{3N}{2\beta} - \frac{1}{2L}\sum_{i,j}(\boldsymbol{R}_i^{(j)} - \overline{\boldsymbol{R}}_i)\cdot\boldsymbol{F}_i^{(j)}, \tag{77}$$

$$E_{\mathrm{est}} = K_{\mathrm{cv}} + \frac{1}{L}\sum_j U(\boldsymbol{R}^{(j)};\boldsymbol{h}), \tag{78}$$

Use physical forces and unwrapped paths, with matching modifications for constraints or removed degrees of freedom [14]. NPT also requires a shared cell, quantum pressure estimator, and correct cell measure [15]:

$$\langle H\rangle_{\mathrm{q}} = \lim_{L\to\infty} \langle E_{\mathrm{est}} + PV\rangle_L, \tag{79}$$

$$C_P^{\mathrm{q}} = \left(\frac{\partial\langle H\rangle_{\mathrm{q}}}{\partial T}\right)_{N,P}. \tag{80}$$

Exclude fictitious kinetic and spring energies from physical enthalpy. A mean-energy estimator's variance is not automatically the quantum enthalpy variance; direct heat-capacity estimators must differentiate the action, normalization, and estimator [14].

Prospective validation Converge beads at every derivative temperature, starting with $\beta\hbar\omega_{\max}/L \ll 1$. Each step costs approximately L force evaluations. An i-PI/PET interface [16] should first recover $L = 1$ classical results, then stable quantum harmonic benchmarks, before loaded-MOF NPT tests. Converge timestep, thermostats, cell size, and sampling as well as beads; validate quantum-displaced configurations against the potential's domain. For H_2O loadings, test O–H vibrations and hydrogen-bond observables alongside heat capacity; high-frequency hydrogen modes can require more beads even near room temperature. Converged PIMD would test the hybrid's missing quantum anharmonic and pressure–volume terms. Do not add a harmonic quantum correction again to PIMD results.

2.5 Graph Theory

Graphs describe atomic neighborhoods and derivative dependencies [17, 21]. Their roles are distinguished in Table 2.

2.5.1 Adjacency, paths, and receptive fields

A graph is described by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of vertices and $\mathcal{E}$ is the set of edges. For instance, suppose we have a set of nodes represented by indices $\mathcal{V} = \{1, 2, 3, 4\}$, the edges correspond to the connections between these nodes, we can represent them as a set of pairs $\mathcal{E} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}$. In the case where there is no directionality in the connections, the graph is said to be undirected and each edge is represented as an unordered pair of vertices (a set). If the connections have a direction, the graph is said to be directed and each edge is represented as an ordered pair of vertices (a tuple). We can easily represent the edges between nodes by constructing the adjacency matrix $\boldsymbol{A}$, where $A_{ij} = 1$ if there is an edge between nodes i and j, and $A_{ij} = 0$ otherwise. The neighborhood of a node i, denoted as $\mathcal{N}_i$, is the set of nodes that are directly connected to i by an edge, which can be defined as $\mathcal{N}_i = \{j : A_{ij} = 1\}$. Atomic graphs are directed periodic multigraphs: messages have directions, and repeated images can connect the same atom labels. Edges indicate interactions, not necessarily chemical bonds. For a simple undirected graph,

$$A_{ij} = \begin{cases} 1, & \{i, j\} \in \mathcal{E}, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{N}_i = \{j : A_{ij} = 1\}. \tag{81}$$

The degree of a node is the number of nodes connected to it. For a node i, the degree is $d_i = |\mathcal{N}_i|$, which allows us to define the degree matrix $\boldsymbol{D} = \mathrm{diag}(d_1, \ldots, d_{|\mathcal{V}|})$. Graph distance counts edges in a shortest path. Its k-hop ball is

$$\mathcal{B}_k(i) = \{j : d_{\mathcal{G}}(i, j) \le k\}. \tag{82}$$

For $1 - 2 - 3 - 4$, $\mathcal{B}_1(1) = \{1, 2\}$ and $\mathcal{B}_2(1) = \{1, 2, 3\}$. Boolean powers of adjacency record reachability (Eq. (128)). With L local message-passing layers, a node depends on its L-hop neighborhood. Atoms influencing the same nonlinear readout can have mixed second derivatives beyond one hop.

Later on, we will apply functions on the graph representation. When using nodes as atoms and edges as interactions, it is important that the ajecency matrix stays invariant with respect to node permuations. For instance, when trying to compute the energy using the graph representation, the result should not depend on the order of the nodes. So $f(\boldsymbol{A}) = f(\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}})$, where $\boldsymbol{P}$ is a permutation matrix. Node outputs must permute consistently; total energy must remain invariant (Eq. (100)). The computation graph instead records operations and directed dependencies. Forward evaluation/AD follows its acyclic order; reverse AD applies the chain rule backward [32].

Table 2: Graph constructions used by the potential and its differentiation.

Graph	Vertices	Meaning of an edge
Interaction	Atoms	Neighbor relation, with periodic-image attributes
Computation	Values or operations	Directed computational dependence
Hessian sparsity	Coordinates (or atom blocks)	Possible off-diagonal curvature coupling
Column intersection	Derivative columns	Possible nonzeros sharing an output row

2.5.2 Graph coloring and sparse automatic differentiation

A proper vertex coloring assigns an integer $c(q) \in \{1, \dots, n_c\}$ to each vertex so that adjacent vertices have different colors. For differentiation, a color identifies input directions that can be processed together. The relevant edges must therefore represent conflicts in derivative recovery, rather than simply atomic neighbor relations.

ASD separates structural preparation from numerical differentiation [33]:

$$\underbrace{\text{detect pattern} \longrightarrow \text{color}}_{\text{preparation}} \longrightarrow \underbrace{\text{evaluate colored products} \longrightarrow \text{decompress}}_{\text{numerical evaluation}}. \tag{83}$$

Preparation can be reused while the pattern remains valid.

From derivatives to a conflict graph. For a function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, let $J_{pq} = \partial f_p / \partial x_q$. A structural sparsity pattern $\boldsymbol{S} \in \{0,1\}^{m \times n}$ satisfies

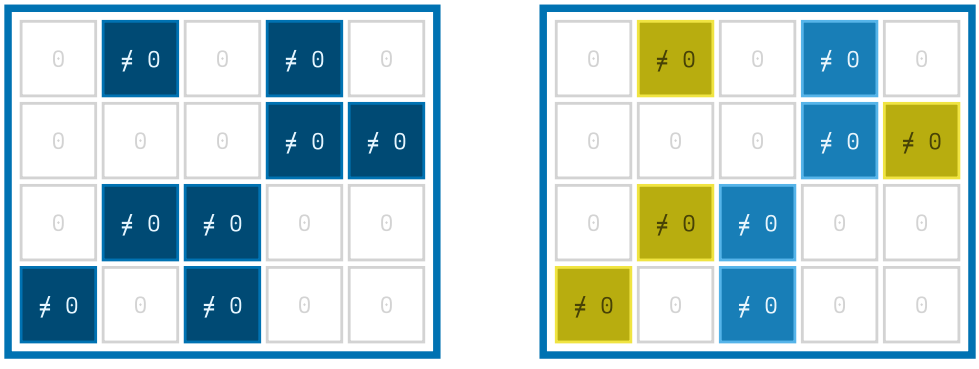
$$S_{pq} = 0 \implies J_{pq} = 0, \qquad \mathcal{R}_q = \{p : S_{pq} = 1\}. \tag{84}$$

Thus $\mathcal{R}_q$ lists the output rows that column q may affect. A numerical zero at one configuration does not necessarily imply a structural zero. Two columns conflict when their supports overlap, so the *column-intersection graph* has one vertex per input coordinate and edges

$$\{q, r\} \in \mathcal{E}_{\text{col}} \iff q \neq r \text{ and } \mathcal{R}_q \cap \mathcal{R}_r \neq \varnothing. \tag{85}$$

Proper coloring of this graph ensures that columns of the same color have disjoint supports, also called structural orthogonality [33].

The pattern can be inferred by propagating input-dependency sets through the computation graph, without evaluating every derivative. For addition, for example, the dependency set of $z = u + v$ is the union of those of u and v. A global pattern covers the input domain; a local pattern may exploit the active branches at one input but requires revalidation [33].



(a) Sparsity pattern (b) Column groups

Figure 2: Structural preparation: permitted entries determine two groups of nonoverlapping columns. Reproduced from Hill, Dalle, and Montoison, Figure 12 [33].

Colored seeds, compression, and recovery. Forward AD evaluates a Jacobian-vector product without first forming $\boldsymbol{J}$. The ordinary seed $\boldsymbol{e}_q$ yields column q; using all n coordinate seeds would therefore require n directional evaluations. For a composition $\boldsymbol{f} = \boldsymbol{h} \circ \boldsymbol{g}$, the chain rule acts on a seed as

$$\boldsymbol{J}_f(\boldsymbol{x})\boldsymbol{v} = \boldsymbol{J}_h(\boldsymbol{g}(\boldsymbol{x}))\big[\boldsymbol{J}_g(\boldsymbol{x})\boldsymbol{v}\big]. \tag{86}$$

AD propagates the direction through these operations without storing the intermediate Jacobian matrices [32]. Construct one seed per color:

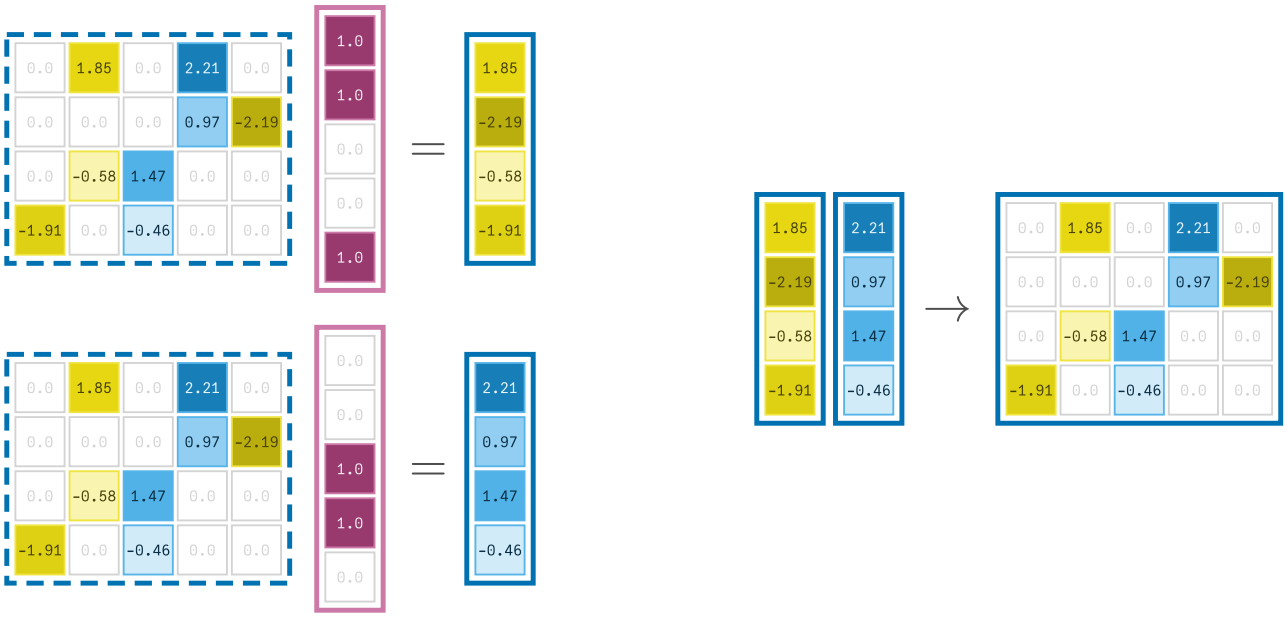
$$\boldsymbol{v}_a = \sum_{q:c(q)=a} \boldsymbol{e}_q, \qquad V_{qa} = \begin{cases} 1, & c(q) = a, \\ 0, & \text{otherwise}, \end{cases} \qquad \boldsymbol{Y} = \boldsymbol{J}\boldsymbol{V}. \tag{87}$$

Each compressed entry is a sum, $Y_{pa} = \sum_{q:c(q)=a} J_{pq}$. Structural orthogonality guarantees that at most one term is permitted to be nonzero in a given row. Consequently,

$$J_{pq} = Y_{p,c(q)} \quad \text{for } S_{pq} = 1. \tag{88}$$

The pattern identifies which column supplied each value. Sparse automatic differentiation (ASD) combines this graph-based preparation with ordinary AD and subsequent decompression; the AD still differentiates the original function. The number of directions falls from n to n_c, rather than computing a dense derivative and discarding its zeros afterward.

Figure 3 illustrates both numerical steps.



(a) Two colored JVPs (b) Decompression

Figure 3: Five columns recovered from two directional evaluations using their known supports. Reproduced from Hill, Dalle, and Montoison, Figure 10 [33].

For the MOF Hessian, take $\boldsymbol{f} = \nabla U$ and $n = m = 3N$. Then $\boldsymbol{J}_f = \boldsymbol{H}$, and a forward derivative of the reverse-mode energy gradient gives $\boldsymbol{H}\boldsymbol{v}_a$ (Eq. (127)). Coloring therefore groups Cartesian coordinates whose Hessian columns can be recovered from the same Hessian-vector product.

Figure 4 connects graph edges to conflicts between derivative columns.

A worked example. Consider the symmetric tridiagonal Hessian

$$\boldsymbol{M} = \begin{pmatrix} a & b & 0 & 0 \\ b & c & d & 0 \\ 0 & d & e & f \\ 0 & 0 & f & g \end{pmatrix}. \tag{89}$$

Its off-diagonal sparsity graph is the chain $1 - 2 - 3 - 4$. Coloring that chain with $(1, 2, 1, 2)$ is proper, but does not permit direct column recovery: columns 1 and 3 both affect row 2, yielding the inseparable sum $b + d$. The column-intersection graph also connects vertices at distance two along the chain. A valid coloring of this larger conflict graph is $(1, 2, 3, 1)$, giving

$$\boldsymbol{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \boldsymbol{M}\boldsymbol{V} = \begin{pmatrix} a & b & 0 \\ b & c & d \\ f & d & e \\ g & 0 & f \end{pmatrix}. \tag{90}$$

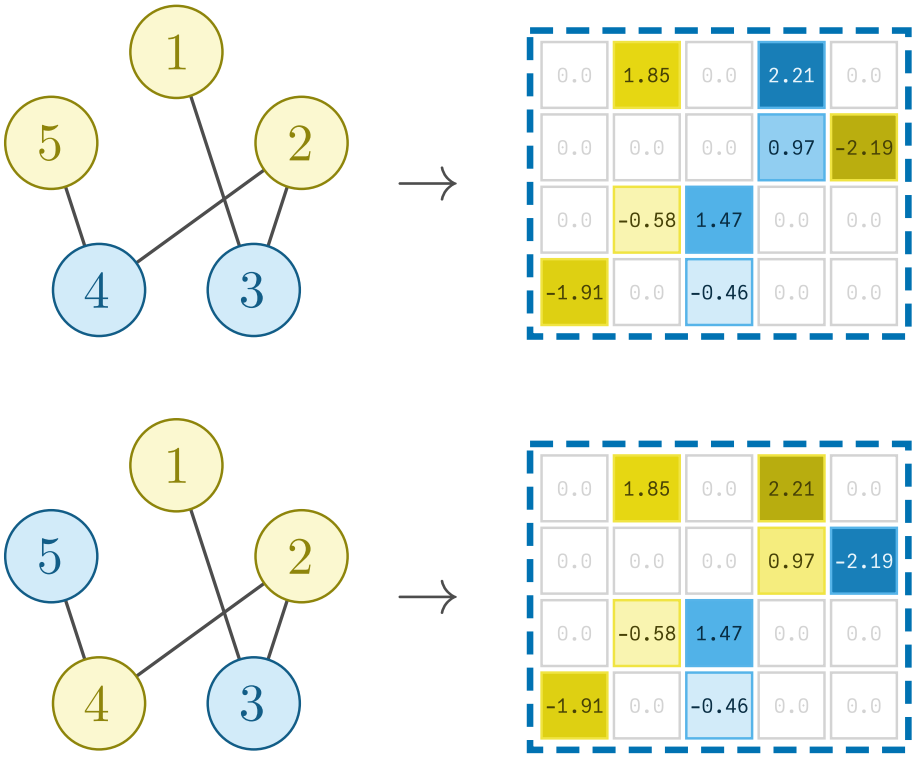


Figure 4: Valid coloring (top) and conflicting same-color columns (bottom). Reproduced from Hill, Dalle, and Montoison, Figures 17 and 19 [33].

The first product combines columns 1 and 4, whose supports are disjoint: rows 1–2 supply column 1 and rows 3–4 supply column 4. Three products recover every entry, whereas dense recovery would use four. For a longer tridiagonal matrix, repeating three colors still suffices, so the benefit grows with matrix size.

Star coloring exploits Hessian symmetry. For a symmetric Hessian, $H_{pq} = H_{qp}$ supplies two possible compressed locations for each off-diagonal value: $Y_{p,c(q)}$ and $Y_{q,c(p)}$. It is sufficient that one of these is uncontaminated. This permits a less restrictive coloring than requiring every pair of column supports to be disjoint [19].

A *star coloring* is a proper coloring of the off-diagonal Hessian sparsity graph in which every path on four distinct vertices uses at least three colors. Equivalently, the subgraph formed by any two colors is a collection of stars. In each star, the center is the hub and its neighbors are spokes. A spoke has only one neighbor with the hub's color, so the hub–spoke entry can be extracted from that spoke's row; symmetry supplies the transposed entry. Proper coloring also keeps diagonal entries separate from off-diagonal contributions [20].

For the same chain, $(1, 2, 1, 3)$ is a star coloring. Its compression is

$$\boldsymbol{W} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{B} = \boldsymbol{M}\boldsymbol{W} = \begin{pmatrix} a & b & 0 \\ b+d & c & 0 \\ e & d & f \\ f & 0 & g \end{pmatrix}. \tag{91}$$

Although $B_{21} = b + d$ cannot be separated directly, $b = B_{12}$ and $d = B_{32}$ are individually available from the opposite orientations. The remaining values are $a = B_{11}$, $c = B_{22}$, $e = B_{31}$, $f = B_{33} = B_{41}$, and $g = B_{43}$. This example uses three colors with either strategy; star coloring can reduce the count further for other symmetric sparsity patterns.

Connection to the project implementation. SADMOF derives an atom-pair pattern from PET's neighborhood graph and the chosen hop reach, then represents each permitted atom pair by a Cartesian 3×3 block. The project passes this pattern to ASDEX for symmetric Hessian coloring and uses forward-over-reverse AD for the colored products (Section 4.4.2). Atom-block indices describe sparsity; the differentiated variables remain the $3N$ Cartesian coordinates. The recovered Hessian is subsequently mass weighted and diagonalized to obtain the harmonic frequencies.

The pattern must include every actual coupling for exact recovery. For example, if the true matrix above also had $M_{24} = M_{42} = \delta$ but the pattern omitted this pair, the direct coloring would still combine columns 1 and 4

and incorrectly recover M_{21} as $b+\delta$. Thus an underestimated pattern can corrupt retained entries as well as omit missing ones. Choosing fewer hops changes the assumed sparsity and requires convergence checks. By contrast, `chunk_size` only controls how many colored directions are evaluated together. A greedy algorithm processes vertices in an ordering and assigns each the smallest admissible color; ordering affects the resulting count. For star coloring, admissibility also enforces the four-vertex-path rule. Greedy coloring need not find the minimum possible number of colors to give correct recovery; extra colors cost work [20].

2.6 Machine-learned interatomic potentials

The potential is what drives the molecular dynamics simulation. However, using first-principles methods like DFT to compute the potential energy surface would be very expensive. On the other hand, using simple potentials like Lennard-Jones or Morse potentials would be too inaccurate for complex systems like MOFs. Machine-learned interatomic potentials (MLIPs) provide a way to approximate the potential energy surface by training neural networks on DFT reference data. This methods represents a good trade-off between accuracy and computational cost, provided it was trained on a representative dataset. It may not be able to give good results for configurations that are far from the training data. This is why there will be an important section about uncertainty quantification in one of the following chapters.

2.6.1 Energy, forces, and atomic graphs

An MLIP approximates the Born–Oppenheimer map $(\boldsymbol{R}, \boldsymbol{h}, \boldsymbol{Z}) \mapsto U$ with fitted parameters θ. A local, extensive decomposition is

$$U_\theta(\boldsymbol{R}, \boldsymbol{h}, \boldsymbol{Z}) = U_{\text{base}}(\boldsymbol{Z}) + \sum_{i=1}^{N} \varepsilon_i(\mathcal{N}_i; \theta), \tag{92}$$

The optional baseline depends only on composition. Atomic energies are model contributions, not unique observables. At bounded neighbor count, evaluation cost scales approximately linearly with atom count. Differentiating the same scalar energy gives conservative forces and a Cartesian Hessian:

$$\boldsymbol{F}_i = -\frac{\partial U_\theta}{\partial \boldsymbol{R}_i}, \tag{93}$$

Atoms are graph vertices with species features. Periodic directed edges $(v, u, \boldsymbol{n})$ carry relative geometry:

$$\boldsymbol{r}_{uv\boldsymbol{n}} = \boldsymbol{R}_v + \boldsymbol{h}\boldsymbol{n} - \boldsymbol{R}_u, \tag{94}$$

$$r_{uv\boldsymbol{n}} = \|\boldsymbol{r}_{uv\boldsymbol{n}}\| < r_{\text{c}}, \tag{95}$$

$$\mathcal{N}_u = \{(v, \boldsymbol{n}) : (v, u, \boldsymbol{n}) \in \mathcal{E}\}. \tag{96}$$

Retaining image labels distinguishes repeated neighbors. Species embeddings and radial edge features provide numerical inputs:

$$\boldsymbol{x}_u^{(0)} = \boldsymbol{E}_Z[Z_u] \in \mathbb{R}^d, \tag{97}$$

$$\boldsymbol{x}_{uv\boldsymbol{n}}^{\text{edge}} = f_{\text{c}}(r_{uv\boldsymbol{n}})\big[\varphi_1(r_{uv\boldsymbol{n}}), \ldots, \varphi_K(r_{uv\boldsymbol{n}})\big], \tag{98}$$

Directional models also use displacement vectors or their learned functions; latent feature vectors need not transform as physical vectors.

Figures 5 and 6 illustrate cutoff weighting. Smooth forces and Hessians require consistent derivatives through attention normalization and readout as well as radial features.

Node-wise F and scalar readout f obey

$$F(\boldsymbol{P}\boldsymbol{X}, \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}}) = \boldsymbol{P}F(\boldsymbol{X}, \boldsymbol{A}), \tag{99}$$

$$f(\boldsymbol{P}\boldsymbol{X}, \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{\mathrm{T}}) = f(\boldsymbol{X}, \boldsymbol{A}). \tag{100}$$

Relabel edge attributes and periodic images consistently.

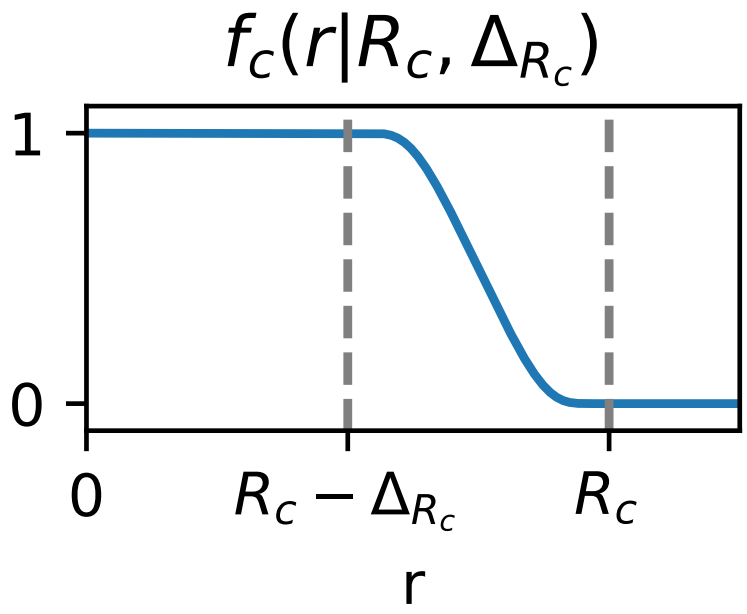


Figure 5: A smooth radial cutoff suppresses a neighbor contribution as its distance approaches the outer cutoff r_c. Smooth decay is needed for continuous energies, forces, and Hessians. Figure from the PET paper [25].

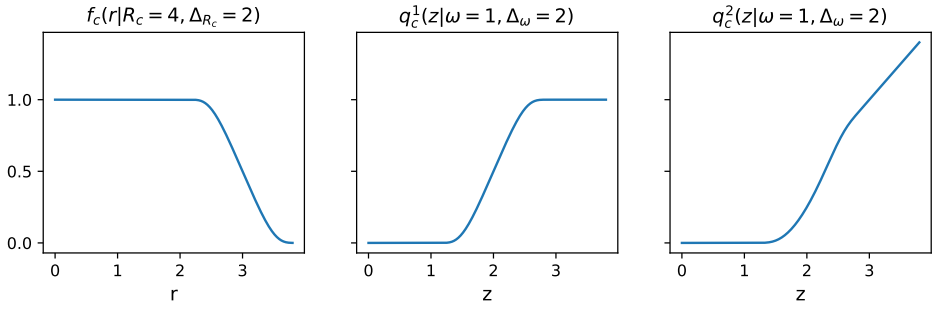


Figure 6: Comparison of radial cutoff functions used to attenuate neighbor contributions near r_c. The value and the derivatives relevant to the model must be treated consistently in every energy contribution. Figure from the PET paper [25].

2.6.2 Message passing

Each layer combines incoming messages with a symmetric aggregation $\bigoplus$ (usually a sum), then updates the node representation [21]:

$$\boldsymbol{h}_u = \phi\left(\boldsymbol{x}_u, \bigoplus_{(v,\boldsymbol{n})\in\mathcal{N}_u} \psi\big(\boldsymbol{x}_u, \boldsymbol{x}_v, \boldsymbol{x}^{\text{edge}}_{uv\boldsymbol{n}}\big)\right). \tag{101}$$

Repeating the update through L layers creates many-body features over an L-hop receptive field. A shared readout $\varepsilon_u = \rho(\boldsymbol{h}_u^{(L)})$ gives the total energy through Eq. (92).

2.6.3 Point Edge Transformer architecture

PET retains distinct directed-edge messages while attention mixes all tokens in one atomic environment (Figure 7) [25]. It underlies PET-MAD [24]. In the original formulation,

$$\boldsymbol{p}_{ij}^{(\ell)} = \mathrm{SiLU}\big(\boldsymbol{W}_r^{(\ell)}\boldsymbol{r}_{ij} + \boldsymbol{b}_r^{(\ell)}\big), \tag{102}$$

$$\boldsymbol{t}_{ij}^{(\ell)} = \mathrm{MLP}_\ell\Big(\boldsymbol{m}_{j\to i}^{(\ell-1)} \| \boldsymbol{E}_n^{(\ell)}[Z_j] \| \boldsymbol{p}_{ij}^{(\ell)}\Big), \tag{103}$$

Here $\|$ concatenates the incoming message, neighbor embedding, and position encoding. The MLP maps $3d_{\mathrm{PET}}$ features to d_{PET}; the first block omits the incoming message. A separate central token is $\boldsymbol{t}_{i0}^{(\ell)} = \boldsymbol{E}_c^{(\ell)}[Z_i]$.

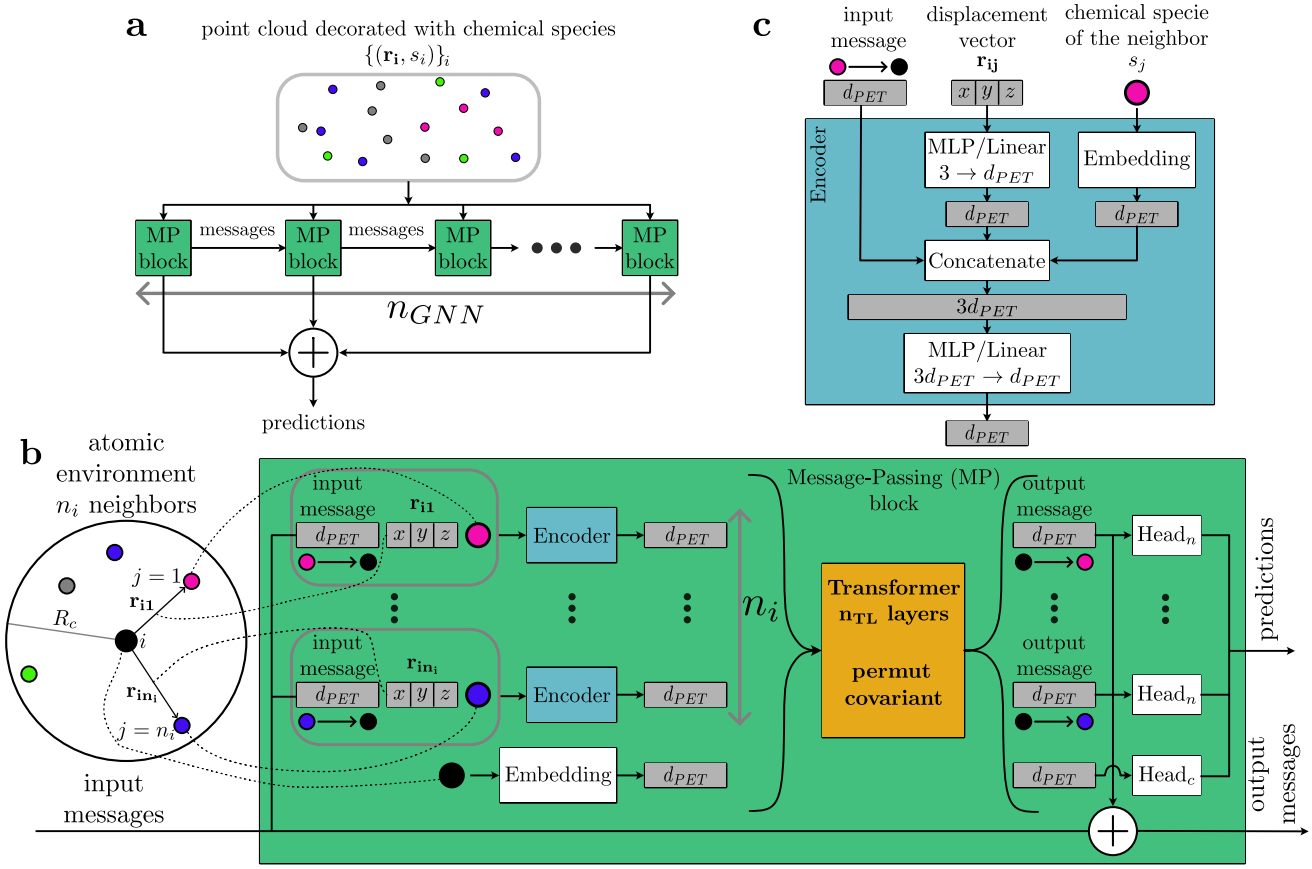


Figure 7: Point Edge Transformer (PET) message-passing architecture. Edge messages associated with a central atom are jointly processed by attention while retaining their individual edge identities for the outgoing update. Figure from the PET paper [25].

For one attention head,

$$\boldsymbol{q}_a = \boldsymbol{W}_Q\boldsymbol{t}_a, \qquad \boldsymbol{k}_b = \boldsymbol{W}_K\boldsymbol{t}_b, \tag{104}$$

$$\boldsymbol{v}_b = \boldsymbol{W}_V\boldsymbol{t}_b, \tag{105}$$

$$\alpha_{ab} = \frac{\exp(\boldsymbol{q}_a^{\mathrm{T}}\boldsymbol{k}_b/\sqrt{d_k})c_b}{\sum_c \exp(\boldsymbol{q}_a^{\mathrm{T}}\boldsymbol{k}_c/\sqrt{d_k})c_c}, \tag{106}$$

$$\boldsymbol{y}_a = \sum_b \alpha_{ab}\boldsymbol{v}_b. \tag{107}$$

With $c_0 = 1$ and $c_j = f_c(r_{ij})$, departing neighbors vanish from both numerator and normalization. Multi-head attention, feed-forward layers, and residual connections preserve neighbor permutation equivariance. Attention layers enrich interactions within one environment; message-passing blocks extend the receptive field. Local attention cost is quadratic in neighbor count, but total cost remains approximately linear in atom count at fixed architecture and bounded neighborhoods.

Outgoing edge tokens update the messages. The original summed-edge readout is

$$U_\theta = \sum_i b_{Z_i} + \sum_{\ell=1}^{L} \sum_i \Big[H_c^{(\ell)}(\boldsymbol{x}_{i0}^{(\ell)}) + \sum_{j \in \mathcal{N}_i} f_c(r_{ij}) H_n^{(\ell)}(\boldsymbol{x}_{ij}^{(\ell)}) \Big], \tag{108}$$

H_c and H_n are learned heads, $\boldsymbol{x}$ are output tokens, and b_Z are species baselines. Every block contributes; cutoff weighting also enters the readout. Edge contributions are learned many-body terms, not pair energies. Project-specific dimensions appear in Section 4.2.

2.6.4 Physical symmetries and ECSE

Scalar energies should respect translation, rotation/reflection, and identical-atom permutation:

$$\begin{aligned} U(\boldsymbol{R}) &= U(\boldsymbol{R} + \boldsymbol{a}) = U(\boldsymbol{Q}\boldsymbol{R}) \\ &= U(\boldsymbol{P}\boldsymbol{R}), \qquad \boldsymbol{Q} \in O(3). \end{aligned} \tag{109}$$

Forces then transform equivariantly, $\boldsymbol{F}_i(\boldsymbol{Q}\boldsymbol{R}) = \boldsymbol{Q}\boldsymbol{F}_i(\boldsymbol{R})$; periodic tests rotate both positions and cell. PET enforces translation and permutation symmetries, but learns rotational behavior through augmentation. Reported rotational errors below prediction errors [24] do not validate this checkpoint on loaded MOF-5.

The optional Equivariant Coordinate System Ensemble (ECSE) imposes exact rotation symmetry through local frames from ordered noncollinear neighbor pairs [25]:

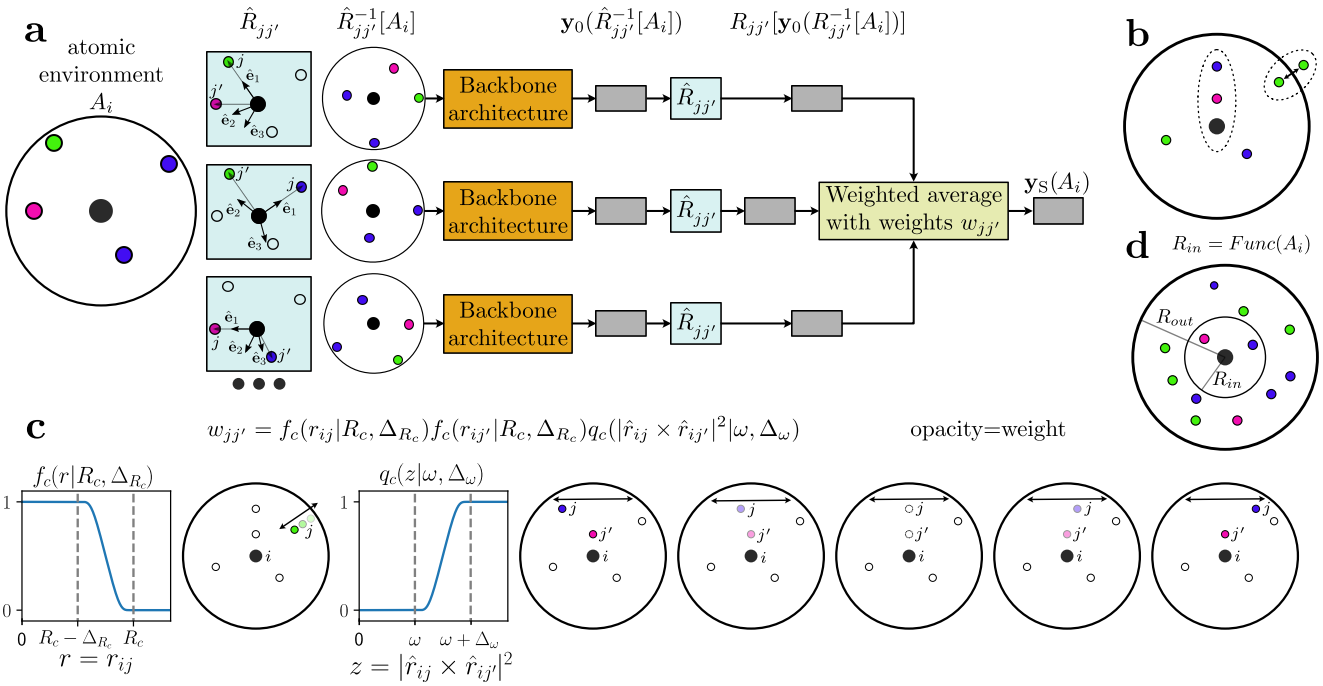


Figure 8: Equivariant Coordinate System Ensemble (ECSE). Local coordinate frames constructed from noncollinear neighbor pairs rotate with the atomic environment; predictions evaluated in these frames are transformed as needed and smoothly averaged. Figure from the PET paper [25].

$$\varepsilon_i^{\mathrm{ECSE}} = \frac{\sum_{j \neq k} w_{ijk}\, \varepsilon_i^0(\boldsymbol{Q}_{ijk}^{\mathrm{T}} \mathcal{N}_i)}{\sum_{j \neq k} w_{ijk}}, \tag{110}$$

$\boldsymbol{Q}_{ijk}^{\mathrm{T}} \mathcal{N}_i$ expresses displacements in a co-rotating frame; vector/tensor outputs are transformed back before averaging. These proper $SO(3)$ rotations alone do not enforce reflection invariance. Smooth radial/angular weights suppress cutoff and nearly collinear frames; fully collinear environments need a fallback. The full frame

and weight dependence must be differentiated for forces and Hessians. Tight frame selection can amplify derivative errors, and the proof-of-principle overhead is roughly 10^3 [25]. ECSE is not enabled here.

2.6.5 Training and validity

A generic energy–force objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\mathrm{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\mathrm{ref}}^{(s)} \right)^2 \right], \tag{111}$$

Stress terms and regularization may be added. PET's original loss uses a different normalization with moving validation errors, species baselines, Adam, and rotational augmentation [25]. PET-MAD is pretrained on a chemically diverse DFT dataset [24].

An MLIP inherits its reference method's approximations and may extrapolate poorly to unfamiliar environments. Hold-out errors, trajectory stability, and uncertainty indicators test different aspects of validity. Curvatures need explicit validation because the usual loss does not constrain Hessians directly; PET's methane benchmarks do not establish accuracy for adsorption in MOF-5.

2.7 Metal–organic frameworks

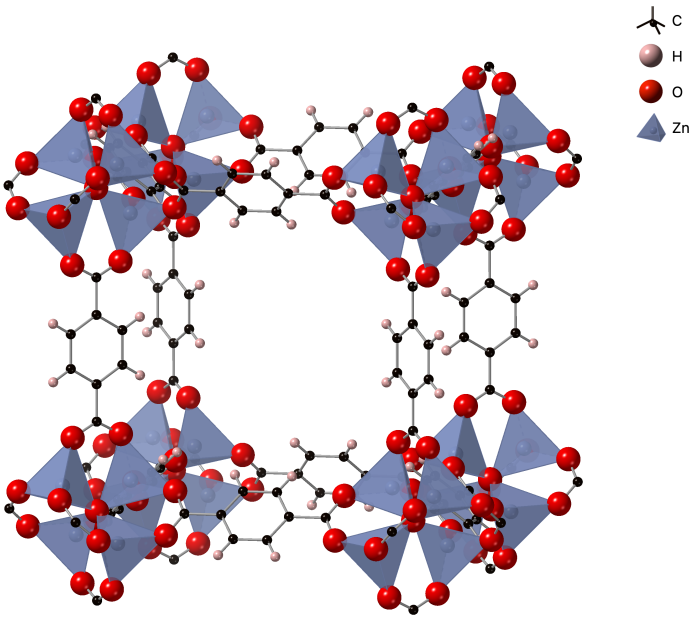


Figure 9: Ball-and-polyhedron representation of the MOF-5 framework. The zinc–oxo clusters are shown as blue polyhedra, connected by benzene-1,4-dicarboxylate linkers. The legend identifies carbon (black), hydrogen (pink), oxygen (red), and zinc (blue). The open spaces between framework components form the pores that host adsorbed molecules.

MOF-5 consists of zinc–oxo nodes joined by benzene-1,4-dicarboxylate linkers (Figure 9). Methane occupies the pores and interacts mainly through weak noncovalent forces. Framework vibration, guest internal modes, and redistribution between adsorption environments all contribute to heat capacity. Loaded-system MD captures the anharmonic motion; its optimized normal modes supply the quantum correction [22].

2.7.1 LLPR model uncertainty and CEA propagation

Finite-trajectory sampling error is distinct from potential uncertainty. LLPR varies the final readout of a fixed PET representation [29]. For central energy $\bar{U} = \boldsymbol{w}_0^{\mathrm{T}} \boldsymbol{f}(\boldsymbol{x})$ and covariance-set features $\boldsymbol{F}$,

$$\begin{aligned} \sigma^2_{\mathrm{LLPR}}(\boldsymbol{x}) = \alpha^2 \boldsymbol{f}(\boldsymbol{x})^{\mathrm{T}} \\ \times \left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} + \lambda \boldsymbol{I}\right)^{-1} \boldsymbol{f}(\boldsymbol{x}). \end{aligned} \tag{112}$$

The regularizer λ stabilizes weakly constrained directions; α is calibrated against energy residuals on a disjoint reference-labeled set using the base potential's electronic-structure definition. Predictions cannot calibrate their own error.

The exported `energy_uncertainty` is an energy standard deviation. Property propagation instead needs persistent, signed energy variations:

$$\begin{aligned} \boldsymbol{w}_i &\sim \mathcal{N}\left[\boldsymbol{w}_0, \alpha^2 \left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} + \lambda \boldsymbol{I}\right)^{-1}\right], \\ U^{(i)}(\boldsymbol{x}) &= \boldsymbol{w}_i^{\mathrm{T}} \boldsymbol{f}(\boldsymbol{x}), \end{aligned} \tag{113}$$

The exported `energy_ensemble` preserves member identity across configurations. Its large-ensemble spread approaches the analytical LLPR value, but cannot capture bias shared by the frozen representation.

Following Imbalzano *et al.* [31], central-trajectory frames supply

$$\Delta U_t^{(i)} = U_t^{(i)} - \bar{U}_t, \tag{114}$$

$$H_t^{(i)} = K_t + U_t^{(i)} + PV_t, \tag{115}$$

$$\left\langle H^{(i)} \right\rangle_i^{\mathrm{direct}} = \frac{\sum_t e^{-\beta \Delta U_t^{(i)}} H_t^{(i)}}{\sum_t e^{-\beta \Delta U_t^{(i)}}}, \tag{116}$$

Extensive energy differences can collapse the exponential weights. The primary large-system estimate uses first-order cumulant expansion (CEA):

$$\begin{aligned} \left\langle H^{(i)} \right\rangle_i^{\mathrm{CEA}} &= \left\langle H^{(i)} \right\rangle_{\bar{U}} \\ &\quad - \beta \,\mathrm{cov}_{\bar{U}}\left(H^{(i)}, \Delta U^{(i)}\right), \\ \beta &= (k_{\mathrm{B}}T)^{-1}. \end{aligned} \tag{117}$$

Direct reweighting remains a diagnostic: $N_{\mathrm{eff}} = 1/\sum_t \tilde{w}_t^2$ should remain adequate, and $\mathrm{var}_t(\beta \Delta U_t^{(i)}) \ll 1$ is needed for controlled first-order CEA.

Average replicas member by member, preserve indices across temperatures, and differentiate each complete enthalpy curve:

$$\begin{aligned} C_{P,\mathrm{cl}}^{(i)}(T) &= \frac{\mathrm{d}\langle H^{(i)} \rangle_i^{\mathrm{CEA}}}{\mathrm{d}T}, \\ \sigma_{\mathrm{LLPR},C_P}(T) &= \left\{ \frac{1}{M-1} \sum_{i=1}^{M} \left[C_{P,\mathrm{cl}}^{(i)}(T) - \overline{C}_{P,\mathrm{cl}}(T)\right]^2 \right\}^{1/2}. \end{aligned} \tag{118}$$

The final property spread is a predictive standard deviation; do not divide it by $\sqrt{M}$.

2.7.2 Gaussian NPT propagation

Direct propagation of shallow ensembles (DPOSE) combines member properties before measuring their spread [30]. The saved comparison uses a Gaussian NPT approximation on central-model trajectories. Define $Q_i = U_i + PV$ and $\Delta U_i = U_i - U_0$. Joint Gaussianity leaves covariance unchanged under exponential reweighting, giving

$$C_{P,i}^{\mathrm{G}} \simeq \frac{f}{2} k_{\mathrm{B}} + \frac{\mathrm{var}_0(Q_i)}{k_{\mathrm{B}} T^2}. \tag{119}$$

Here $f = 3N - 3$. The central-model fluctuation estimator is exact under the usual equilibrium assumptions; using central-trajectory variances for other members needs Gaussian closure. Pair member indices with harmonic corrections.

The main Gaussian plots replace only the propagated model spread around the central enthalpy-derivative curve. The separate variance-based hybrid also replaces the central classical value and estimates sampling error from squared configurational-enthalpy fluctuations. Neither guarantees smaller or better calibrated uncertainty.

2.7.3 Classical versus quantum vibrations

Classical harmonic modes contribute k_B each at every temperature. Quantum occupation decreases when $\hbar\omega \geq k_B T$ (Eq. (141)). Longer classical trajectories cannot recover that suppression.

3 Reference evidence and the hybrid approximation

Kapil *et al.* tested empty MOF-5 and 50-, 100-, and 150-methane cells [22]. Their hybrid strategy motivates this project; its accuracy on PET remains to be established.

3.1 Differences from the reference calculation

The reference used QuickFF [26]: B3LYP cluster-based bonded terms, MBIS charges from periodic PBE, Gaussian electrostatics, and MM3 Buckingham interactions with a 15 Å cutoff and tail corrections. RASPA insertion and canonical Monte Carlo preceded MD. Here PET supplies all energies, forces, and Hessians after independent random guest placement.

Both workflows differentiate neighboring equilibrium enthalpies:

$$C_P(T) \simeq \frac{\langle H\rangle_{T+\Delta T} - \langle H\rangle_{T-\Delta T}}{2\Delta T}, \qquad \Delta T = 25\,\mathrm{K}, \tag{120}$$

The reference used optimized Yaff Hessians and TAMkin modes, compared QuickFF and UFF4MOF for the empty host, and optimized five Monte Carlo configurations for the 100-methane harmonic calculation.

3.2 Physical conclusions

The reference found nearly harmonic, quantum-mechanical empty MOF-5 (roughly 0.7 J/g/K near room temperature). Classical occupations overestimate its heat capacity. Methane introduces anharmonic host–guest motion and a loading-dependent minimum near 200 K, associated with changing attractive interactions as guests become mobile. The combination

$$C \approx C_{\mathrm{qn}}^{\mathrm{har}} - C_{\mathrm{cl}}^{\mathrm{har}} + C_{\mathrm{cl}}^{\mathrm{anh}} \tag{121}$$

reproduced the minimum qualitatively and was useful above about 200 K in their 100-methane test. A loaded-system spectrum is essential: the empty Hessian omits guest modes and host–guest coupling. Equation (145) gives the operational PET estimator.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The conventional host has 424 atoms and cell parameter 25.866 Å. The campaign module `mof_heat_capacity.protocols.loaded` inserts centered methane molecules with uniform quaternion rotations and fractional positions, rejecting periodic atom–atom overlaps. Independent insertion and velocity seeds define replicas. Random placement prevents severe contacts but is not an equilibrium adsorption distribution; subsequent equilibration must be validated.

4.2 PET model and implementation

The supplied PET-MAD and PET-SOL conversions share the architecture in Table 3; identical settings do not imply identical weights or training data. The original PET default used three attention layers per block, rather than one here.

The installed PyTorch implementation (`metatrain` 2026.3.1) embeds $(x_{ij}, y_{ij}, z_{ij}, r_{ij})$ and initializes edge messages with species embeddings. Central features contract to token width for attention and expand afterward. Attention-head width is $256/8 = 32$; `d_head=256` denotes readout width. Neighbor masking and log-cutoff logits implement Eq. (107). Reverse-neighbor routing and averaging with old messages update edge states.

The repository exports checkpoints with `upet.save_upet` for LAMMPS/metatomic MD, and converts matching weights to PET-JAX for SADMOF Hessians. Setup pins PET-JAX to commit `ed0add4`; conversion does not retrain. Composition baselines have zero Cartesian derivatives; energy scales must carry through forces and Hessians.

Compare energies and forces across export and conversion, including adaptive-cutoff derivative conventions. The sparse path stops those derivatives (Section 4.4.2). NPT additionally requires validated stress output on representative loaded-MOF structures; matching filenames or capability flags alone are insufficient [24].

Table 3: Architecture settings recorded in the local PET-MAD and PET-SOL PET-JAX metadata. The species count is embedding capacity, not a measure of training coverage.

Setting	Value
Outer cutoff	8.0 Å
Cutoff switching width	0.5 Å
Adaptive neighbor-selection setting	40
Message-passing blocks	3
Attention layers per block	1
Token width, d_{PET}	256
Attention heads	8
Transformer feed-forward width	512
Node feature width, `d_node`	1024
Readout feature width, `d_head`	256
Species-embedding capacity	102

4.3 Classical NPT molecular dynamics

Only loaded MOF-5 is propagated. LAMMPS/metatomic uses a flexible triclinic cell with isotropic 1 bar target pressure and a Nosé–Hoover-chain/MTTK NPT fix. The campaign settings are

Table 4: Configured NPT campaign. Equilibration and convergence are assessed from observables, not assumed from these durations.

Parameter	Value
Thermostat chain length	3
Thermostat / barostat damping	100 / 1000 fs
Timestep / total steps	0.5 fs / 10^6
Trajectory / discarded interval	500 / 100 ps
Save interval	100 steps (0.05 ps)
Models / methane loadings	PET-MAD, PET-SOL / 50, 100, 150
MD grid	175–425 K, every 25 K
Hybrid reporting grid	200–400 K, every 25 K
Available replicas	One per state point

Production means use physical enthalpy,

$$\langle H \rangle_T = \langle E + PV \rangle_T \,, \tag{122}$$

then replica averaging, where available, and centered differentiation with neighbors separated by 50 K. This term retains sampled anharmonic motion. Independent replicas, longer production, and temperature-spacing convergence remain outstanding (Section 5.2).

4.4 Harmonic approximation and Cartesian Hessian

Let $\boldsymbol{R}_0$ be a stationary reference structure and $\boldsymbol{u} = \boldsymbol{R} - \boldsymbol{R}_0$. Expanding the potential energy gives

$$\begin{aligned} U(\boldsymbol{R}_0 + \boldsymbol{u}) = U(\boldsymbol{R}_0) + \underbrace{\nabla U(\boldsymbol{R}_0) \cdot \boldsymbol{u}}_{=0 \text{ at a minimum}} \\ + \frac{1}{2} \boldsymbol{u}^{\mathrm{T}} \boldsymbol{H} \boldsymbol{u} + \mathcal{O}(u^3), \end{aligned} \tag{123}$$

with Cartesian Hessian

$$H_{i\alpha,j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = -\frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \tag{124}$$

For N atoms, $\boldsymbol{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772.

4.4.1 Automatic differentiation

At fixed parameters, a neural-network energy composes differentiable operations:

$$\boldsymbol{z}^{(0)} = \boldsymbol{R}, \qquad \boldsymbol{z}^{(\ell+1)} = \phi_\ell(\boldsymbol{z}^{(\ell)}; \theta), \qquad U_\theta = \rho(\boldsymbol{z}^{(L)}). \tag{125}$$

AD applies their chain rules without a finite-displacement step [32]. It differentiates the implemented potential, up to floating-point error; it does not improve the potential's accuracy. For $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$,

$$(\boldsymbol{J}_f)_{pq} = \frac{\partial f_p}{\partial x_q}, \qquad \boldsymbol{J}_f \in \mathbb{R}^{m\times n}. \tag{126}$$

Forward AD evaluates $\boldsymbol{J}_f\boldsymbol{v}$ (JVP); reverse AD evaluates $\boldsymbol{w}^{\mathrm{T}}\boldsymbol{J}_f$ (VJP). Dense recovery requires n forward or m reverse seeds, without storing layer Jacobians [33]. A single reverse sweep yields ∇U and hence all forces. Forward-over-reverse evaluates

$$\boldsymbol{H}\boldsymbol{v} = \frac{\mathrm{d}}{\mathrm{d}\epsilon}\nabla_{\boldsymbol{R}} U_\theta(\boldsymbol{R} + \epsilon\boldsymbol{v})\bigg|_{\epsilon=0}. \tag{127}$$

Dense Hessian recovery needs $3N$ such products. Sparse reconstruction reduces the seed count; cutoff smoothness and neighbor-selection conventions still control which function is differentiated.

4.4.2 Sparse Hessian reconstruction with SADMOF

SADMOF constructs an atom-pair pattern from PET's selected-neighbor graph, then expands each pair into a Cartesian 3×3 block [34]. For L message-passing layers, the maximum Hessian reach is $K = 2L$ for node readout and $K = 2L+1$ for PET edge readout. The paper's MACE-MP-0 medium, PET-XS, and PET-S reaches are four, five, and seven hops. Boolean adjacency gives

$$\boldsymbol{P}^{(k)} = \boldsymbol{I} \vee \boldsymbol{A} \vee \boldsymbol{A}^2 \vee \cdots \vee \boldsymbol{A}^k, \tag{128}$$

At $k = K$ the pattern contains all couplings under the chosen derivative convention; $k < K$ truncates it. Additional nonlocal terms need separate analysis. Structural sparsity requires

$$S_{pq} = 0 \quad \Longrightarrow \quad H_{pq} = 0 \tag{129}$$

An overestimate costs entries and colors. An underestimate can both discard curvature and contaminate retained entries. Mixed second derivatives depend on nonlinear readout, not just direct interaction edges.

As developed in Section 2.5.2, direct column recovery connects columns with overlapping supports:

$$(q, r) \in \mathcal{E}_S \quad \Longleftrightarrow \quad \exists p :\ S_{pq} = S_{pr} = 1. \tag{130}$$

A proper coloring makes same-color columns structurally orthogonal. Greedy heuristics need not minimize color count to recover exact entries [33]. Define one seed per color class,

$$\boldsymbol{v}_a = \sum_{q\in\mathcal{C}_a} \boldsymbol{e}_q, \qquad \boldsymbol{H}\boldsymbol{v}_a = \sum_{q\in\mathcal{C}_a} \boldsymbol{H}\boldsymbol{e}_q. \tag{131}$$

and collect seeds in $\boldsymbol{V} \in \{0,1\}^{3N\times n_c}$:

$$\boldsymbol{Y} = \boldsymbol{H}\boldsymbol{V}, \qquad Y_{pa} = \sum_{q:c(q)=a} H_{pq}. \tag{132}$$

Disjoint supports permit

$$H_{pq} = Y_{p,c(q)} \qquad \text{when } S_{pq} = 1, \tag{133}$$

with zeros outside the pattern.

ASDEX instead star-colors the symmetric Hessian graph. For each pair $\{p, q\}$, a hub–spoke orientation selects an uncontaminated compressed location and writes it to both H_{pq} and H_{qp}. This can reduce colors [33, 34]. Atom-block coloring can be lifted to coordinate colors $3c_i + d$, $d \in \{0, 1, 2\}$; availability depends on the installed ASDEX implementation.

The pipeline is summarized by

$$\boldsymbol{S} \xrightarrow{\text{color}} \boldsymbol{V} \xrightarrow{\text{HVPs}} \boldsymbol{Y} = \boldsymbol{H}\boldsymbol{V} \xrightarrow{\text{recover}} \boldsymbol{H}, \qquad \text{ideal AD reduction} = \frac{3N}{n_c}. \tag{134}$$

Pattern setup, compilation, decompression, and GPU utilization limit the wall-time gain. Reuse a coloring only while its structural pattern remains valid.

The wrapper uses `atoms_to_inputs`, `sparsity_pattern`, `asdex.hessian_coloring_from_sparsity`, `get_energy_fn`, `get_hessian_fn`, and `jax.jit`. It densifies the sparse result and removes padding before eigendecomposition; only the AD stage exploits sparsity.

`no_shadow=True` stops adaptive-neighbor-selection derivatives and uses the selected graph. Shadow derivatives introduce extra couplings; `shadow=true` is rejected in this sparse path. Check consistency with MD forces. With three PET blocks, the paper derives $K = 7$, so the configured `hops=3` is an approximation requiring convergence.

Truncated ASD does not simply zero distant blocks of a dense Hessian: omitted couplings can share seeds with retained entries. For Cartesian mask $\boldsymbol{M}_k$ and reconstructed $\widetilde{\boldsymbol{H}}_k$ before acoustic treatment,

$$\Delta \boldsymbol{H}_{\mathrm{disc}} = -(\mathbf{1} - \boldsymbol{M}_k) \odot \boldsymbol{H}, \tag{135}$$

$$\Delta \boldsymbol{H}_{\mathrm{cont}} = \boldsymbol{M}_k \odot (\widetilde{\boldsymbol{H}}_k - \boldsymbol{H}), \tag{136}$$

$$\|\widetilde{\boldsymbol{H}}_k - \boldsymbol{H}\|_F^2 = \|\Delta \boldsymbol{H}_{\mathrm{disc}}\|_F^2 + \|\Delta \boldsymbol{H}_{\mathrm{cont}}\|_F^2. \tag{137}$$

Here $\odot$ is entrywise multiplication. The paper finds contamination tracking discarded norm down to roundoff; this is empirical, not a general bound. Acoustic-sum-rule enforcement does not remove either error.

At 0.1% heat-capacity tolerance at 300 K, most paper benchmarks converge at two hops for MACE, three for PET-S, and four for PET-XS, with slower zeolite cases [34]. Loaded MOF-5 requires its own comparison against exact-pattern or feasible dense AD, inspecting signed modes and the correction across temperatures. Convergence can be nonmonotonic near mode thresholds. Periodic image folding also requires cell/supercell convergence.

Table 5: Numerical controls for Hessian evaluation. Template settings are not established convergence values; archived results use float64.

Control	Effect	Template
`hops`	Changes the assumed sparsity approximation	3
`chunk_size`	Batches colored directions; affects memory/throughput	1
`remat`	Recomputes intermediates to reduce memory	true
`dtype`	Sets floating-point precision	float32

Keep separately tagged convergence outputs, including higher precision and less truncated references, especially for low-frequency modes.

4.5 Normal modes and quantum harmonic heat capacity

Symmetrize the Hessian, enforce the acoustic sum rule, and mass weight:

$$\sum_j H_{i\alpha,j\beta} = 0, \tag{138}$$

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \qquad \boldsymbol{D}\boldsymbol{e}_k = \omega_k^2 \boldsymbol{e}_k. \tag{139}$$

Uniform translations should have zero restoring force. Unlike the upstream SADMOF observable, which maps negative eigenvalues to zero, the local calculation preserves signed imaginary frequencies. Strict hybrid assembly rejects unexpected imaginary modes and allows at most three translational near-zero modes. Exploratory outputs instead discard modes at or below the threshold (Section 5.5).

For retained positive modes, use the oscillator heat capacity derived in Section 2.3.3:

$$x_k = \frac{\hbar \omega_k}{k_{\mathrm{B}} T} = \frac{h_{\mathrm{P}} c \tilde{\nu}_k}{k_{\mathrm{B}} T}, \tag{140}$$

$$C_{V,k} = k_{\mathrm{B}} \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \tag{141}$$

$$c_V(T) = \frac{1}{M_{\mathrm{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \tag{142}$$

$\tilde{\nu}_k$ is wavenumber, and M_{cell} includes guests. Using cell mass in grams and mode heat capacity in J/K gives J/g/K. Equivalently use $N_{\mathrm{A}}/M_{\mathrm{molar}}$. One spectrum supplies the configured 100–500 K curve every 10 K; temperature changes populations, assuming fixed force constants.

4.6 Optimized minima and the hybrid correction

Relax the equilibrated empty structure directly at fixed cell for a separate reference. For loaded systems, quench representative independent replicas with the same MLIP and accept only force-converged, stable minima. Raw thermal frames are not harmonic references. Current outputs use one geometry per model/loading from 400 K.

For N_m accepted minima,

$$\overline{\Delta c^{\mathrm{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \Big[c_{\mathrm{qn},s}^{\mathrm{har}}(T) - c_{\mathrm{cl},s}^{\mathrm{har}} \Big], \tag{143}$$

$$s^2_{\Delta c}(T) = \frac{1}{N_m - 1}\sum_{s=1}^{N_m}\Big[\Delta c_s^{\mathrm{har}}(T) - \overline{\Delta c^{\mathrm{har}}}(T)\Big]^2. \tag{144}$$

Use the same retained modes for quantum and classical terms, with k_B per classical mode. Mixed host–guest motion requires the complete loaded spectrum.

$$\begin{aligned} c_P^{\mathrm{approx}}(T) &= c_{P,\mathrm{cl}}^{\mathrm{anh}}(T) + \overline{\Delta c^{\mathrm{har}}}(T), \\ c_{P,\mathrm{cl}}^{\mathrm{anh}}(T) &= \frac{1}{M_{\mathrm{cell}}}\frac{\mathrm{d}\langle H\rangle_{\mathrm{cl}}}{\mathrm{d}T}. \end{aligned} \tag{145}$$

This combines an anharmonic classical C_P with a harmonic C_V correction, assuming a small quantum pressure–volume term. Validation below about 200 K remains limited.

Persistent LLPR members must contribute both terms before their spread is taken:

$$\sigma^2_{\mathrm{model}} = \sigma^2_{\mathrm{cl,model}} + \sigma^2_{\mathrm{har,model}} + 2\,\mathrm{cov}_i(c_{P,\mathrm{cl}}^{(i)}, \Delta c_{\mathrm{har}}^{(i)}), \tag{146}$$

$$u^2_{\mathrm{hyb}} = \sigma^2_{\mathrm{MD}} + \sigma^2_{\mathrm{har,samp}} + \sigma^2_{\mathrm{model}}. \tag{147}$$

The covariance is across matched model members. σ_{MD} is the sampling error of the central enthalpy derivative; $\sigma_{\mathrm{har,samp}}$ comes from independently quenched structures. Quadrature assumes independence of the sampling and model contributions.

Current single-replica analysis perturbs enthalpy means with Gaussian draws using autocorrelation-based errors and recomputes the derivative. It supplies within-trajectory error, not a block bootstrap or between-replica variance. Bands omit the unestimated harmonic sampling term: $u_{\mathrm{hyb}} = (\sigma^2_{\mathrm{MD}} + \sigma^2_{\mathrm{model}})^{1/2}$. LLPR Hessians use central-model geometries without member-specific relaxation.

5 Results and discussion

5.1 Available calculations and status

The campaign contains 66 trajectories: PET-MAD (`pet-mad-1.5-s-40nn`) and PET-SOL (`pet-sol-s-best`), three loadings, eleven temperatures, and one replica per state point. Each 500 ps run contributes 8001 frames after discarding 100 ps: 33 ns simulated and 26.4 ns analyzed in total. Hybrid curves span 200–400 K; 175/425 K support centered endpoint derivatives. Each curve reuses one loaded 400 K reference spectrum.

All estimates are *exploratory*: every loaded spectrum has imaginary modes, and four of six relaxations failed their convergence criterion. Discarding unstable modes does not satisfy the strict acceptance protocol. Bands quantify selected sampling and model errors, excluding unstable-mode bias and unestimated variation across minima.

Trajectory diagnostics come from per-run `summary.json`; central values and relaxation diagnostics come from `heat-capacity.exploratory-discard-imaginary` CSV/JSON archives. Reported bands use refreshed stride-1 evaluation of all 8001 frames with 64 persistent LLPR members, rather than older per-loading uncertainty columns. Figure sources are recorded in `results-provenance.json`.

5.2 Thermodynamic sampling and equilibration

Mean temperatures stay within 0.315 K of target. PET-MAD/100 at 300 K gives 299.854 ± 8.031 K (mean and standard deviation), close to the equipartition width $T\sqrt{2/(3N-3)} = 8.058$ K. This checks kinetic sampling, not configurational ergodicity. Pressure fluctuations span 0.80–1.51 $\times 10^3$ bar with mean errors of 9–18 bar; PET-MAD/100 gives 4.65 ± 13.11 bar, compatible with the 1 bar target. Physical enthalpy always uses imposed pressure.

Enthalpy increases smoothly (Figure 10). Archived enthalpy correlation times of 0.095–0.101 ps imply 1980–2115 effective samples and mean errors of 0.015–0.046 eV. These short-time estimates do not exclude slow drift.

Half-trajectory discrepancies flag 33 of 66 runs with $|z_H| > 2$, where

$$z_H = \frac{\overline{H}_{\mathrm{second}} - \overline{H}_{\mathrm{first}}}{\sqrt{\mathrm{SE}^2_{\mathrm{first}} + \mathrm{SE}^2_{\mathrm{second}}}}. \tag{148}$$

Examples are $z_H = 7.47$ for PET-MAD/150 at 175 K and 4.57 for PET-SOL/150 at 300 K. Some RMS-force correlation times reach 26.1 ps, leaving fewer than eight effective samples.

In Figure 11, the accumulated mean differs from the final mean by about −0.30 eV after 100 ps and −0.15 eV after 200 ps. A large model spread does not absorb a temperature-dependent sampling bias. Longer independent runs and sensitivity to discarded intervals are needed to validate the reported small MD errors.

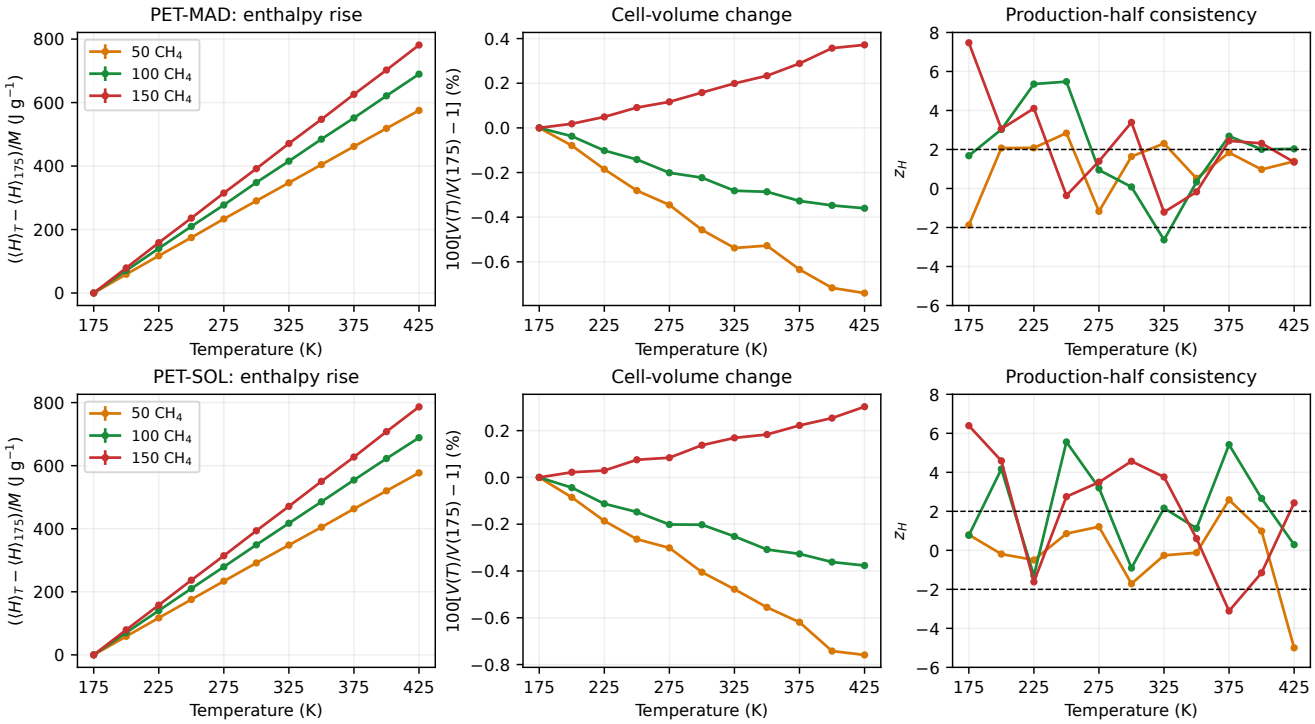


Figure 10: Thermodynamic diagnostics from all 66 trajectory summaries. Left: mass-specific enthalpy increase relative to the same model and loading at 175 K; the baseline is a display offset, and the small bars show the pointwise enthalpy standard errors without propagating the shared baseline error. Middle: fractional change of mean cell volume relative to 175 K. Right: the archived difference between production-half mean enthalpies divided by their combined standard error, z_H; dashed lines mark ±2. These are diagnostics of the individual runs, not replica averages.

5.3 Cell response, framework motion, and methane mobility

Both models predict contraction at 50/100 guests and expansion at 150 over 175–425 K (Figure 10). PET-MAD volumes change from 17146.8 to 17019.9, 17170.2 to 17108.4, and 17210.4 to 17 274.4 Å^3, respectively. Across models the changes are about −0.75%, −0.37%, and +0.3–0.4%. Densities at 300 K are about 0.677, 0.752, and 0.825 g/cm^3, largely reflecting added guest mass. Because $C_P - C_V \propto \alpha_V^2$, contraction does not imply a negative heat-capacity difference; compressibility is also needed to quantify it.

At 300 K, framework RMSDs are 0.659–0.725 Å, while tracked bond-length changes are only 0.055–0.059 Å. Bounded RMSD supports collective motion without gross collapse over the analyzed interval, but tracking a reference neighbor set cannot rule out all bond changes.

Methane 400 ps MSDs at 300 K decrease with loading: 8.9–9.2, 5.1–5.6, and 3.1–3.4 × 10^3 Å^2 for 50, 100, and 150 molecules, respectively. Guest radial distributions peak near 4 Å. Mobility and crowding support the need for anharmonic MD; a single harmonic geometry cannot describe diffusion. Endpoint MSDs alone do not establish diffusion coefficients without a verified diffusive regime, cell-motion control, and replica statistics.

5.4 Classical, harmonic, and hybrid heat capacities

Table 6 resolves the 300 K contributions. The negative harmonic correction removes 59–63% of the classical term, as quantum occupations suppress high-frequency vibrations. Its increase in magnitude with loading reflects added methane and mixed host–guest modes, requiring loaded rather than empty spectra.

Room-temperature hybrid values rise from approximately 0.93 to 1.07–1.08 and 1.16 J/g/K with loading. Model differences of 0.003, 0.011, and 0.001 J/g/K are below propagated bands and cannot distinguish predictive accuracy.

The curves generally rise over 200–400 K (Figures 13–14). For PET-MAD/50, the classical term decreases from 2.347 to 2.269 J/g/K, while the correction becomes less negative, from −1.551 to −1.164 J g^{-1} K^{-1}. Thus the upward hybrid slope need not indicate increasing classical anharmonicity. Adjacent loading bands overlap; significance requires paired propagation of their difference. The grid does not bracket the reference low-temperature minimum, and small local bends remain unresolved within uncertainty.

Empty-framework 300 K values are 0.786 ± 0.021 (PET-MAD) and 0.766 ± 0.027 J g^{-1} K^{-1} (PET-SOL). These are framework-mass-normalized quantum harmonic c_V values with LLPR spread, not hybrid c_P or incremental adsorption heat capacities.

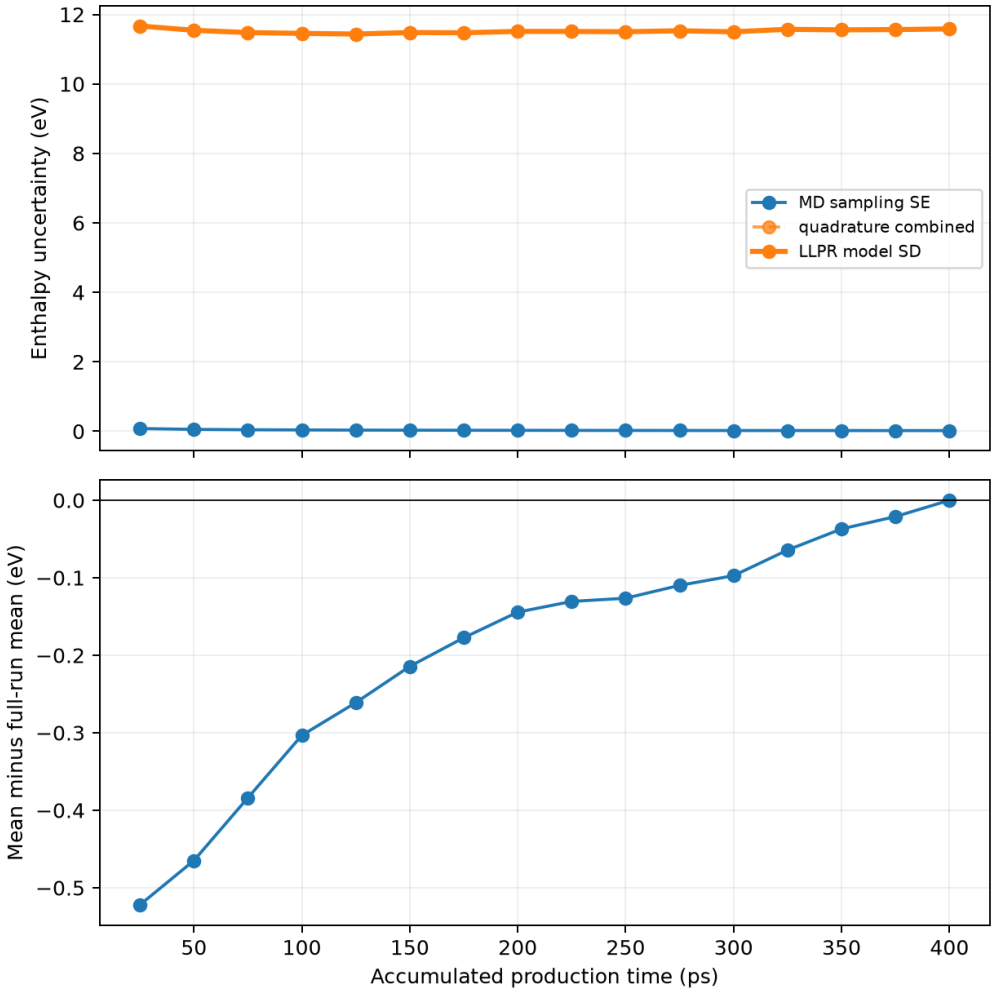


Figure 11: Archived enthalpy convergence diagnostic for PET-MAD with 150 methane molecules at 175 K. The upper panel separates MD sampling error from the LLPR spread of the *enthalpy*, in eV. The lower panel shows the accumulated mean relative to the full-production mean. Its endpoint is zero by construction and is not evidence of equilibration.

Table 6: Heat capacities and uncertainty estimates at 300 K, all in J/g/K and normalized by the total mass of each loaded cell. c_{cl} is the central enthalpy derivative, Δc_{har} the retained-mode harmonic correction, and c_{hyb} their sum. u_{CEA} and u_{G} use different paired model spreads around the *same* central value and MD standard error σ_{MD}. The final column is the separate variance-based central hybrid estimate. None of these uncertainties includes the bias from discarded imaginary modes or an estimated spread across minima.

Model	$N_{\mathrm{CH_4}}$	c_{cl}	Δc_{har}	c_{hyb}	σ_{MD}	u_{CEA}	u_{G}	$c_{\mathrm{hyb}}^{\mathrm{var}}$
PET-MAD	50	2.281	-1.351	0.929	0.010	0.068	0.127	0.926
PET-MAD	100	2.762	-1.682	1.080	0.010	0.123	0.169	1.044
PET-MAD	150	3.122	-1.957	1.165	0.010	0.113	0.214	1.202
PET-SOL	50	2.289	-1.363	0.926	0.010	0.087	0.150	0.940
PET-SOL	100	2.762	-1.693	1.069	0.010	0.153	0.225	1.024
PET-SOL	150	3.134	-1.969	1.166	0.010	0.103	0.306	1.230

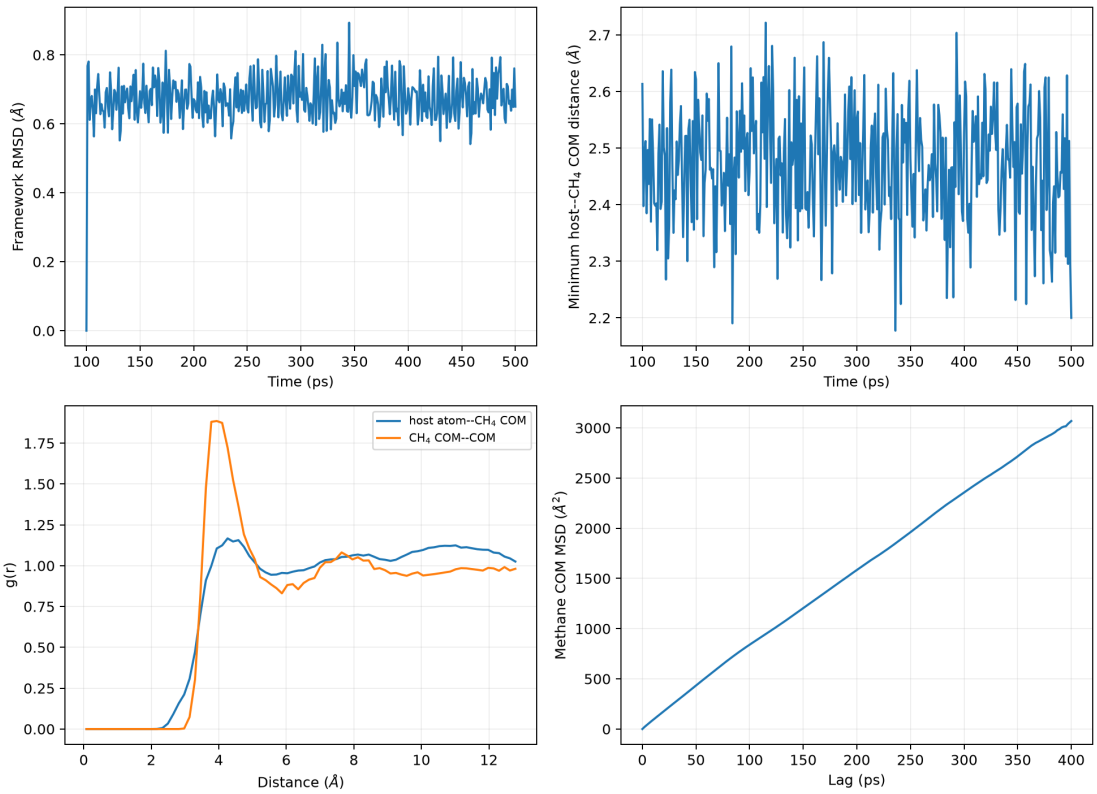


Figure 12: Structural diagnostics for PET-SOL with 150 methane molecules at 300 K: framework RMSD, nearest host-atom–methane center-of-mass distance, radial distributions, and methane center-of-mass mean-squared displacement (MSD). The first analyzed frame at 100 ps defines the framework reference, hence the initial zero RMSD. Long-lag MSD values use fewer time origins than short-lag values.

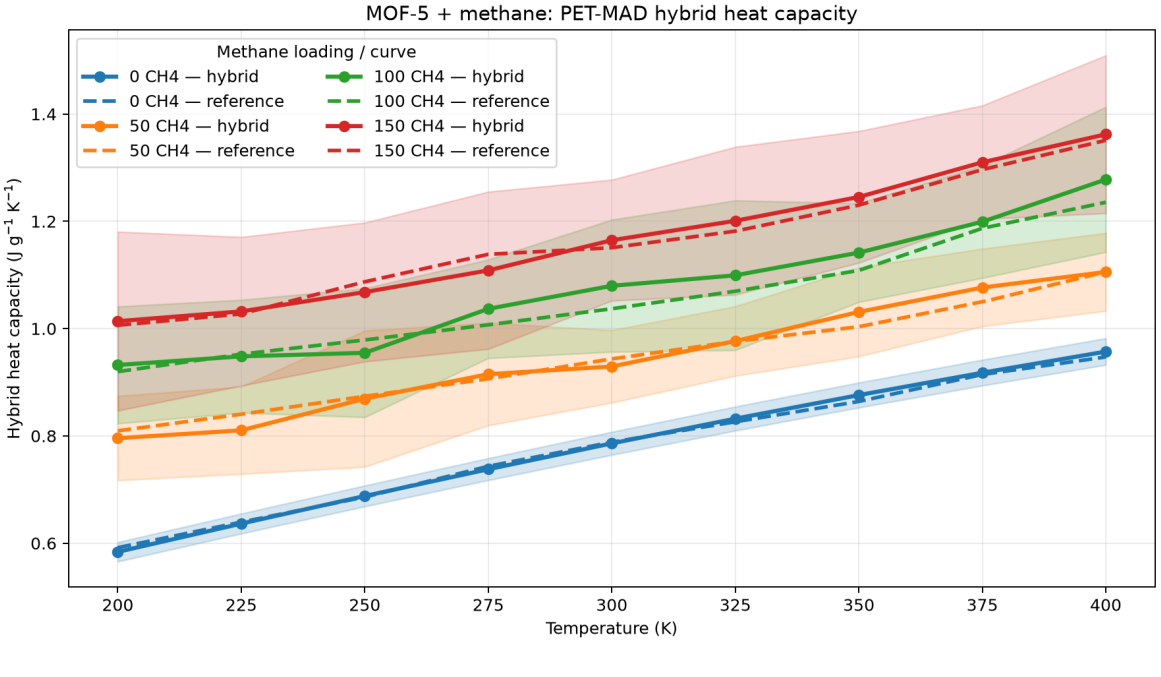


Figure 13: Supplied PET-MAD loading comparison. Loaded solid curves are central classical enthalpy derivatives plus retained-mode harmonic quantum corrections. Their shading is $\pm u_{\mathrm{CEA}}$, combining MD sampling error with correlated classical–harmonic LLPR spread. Despite the common "hybrid" legend, the zero-loading curve is a quantum-harmonic c_V calculation only, with a harmonic LLPR band; no empty-system MD term was computed. Dashed curves are the supplied reference CSVs, whose provenance is discussed in the text.

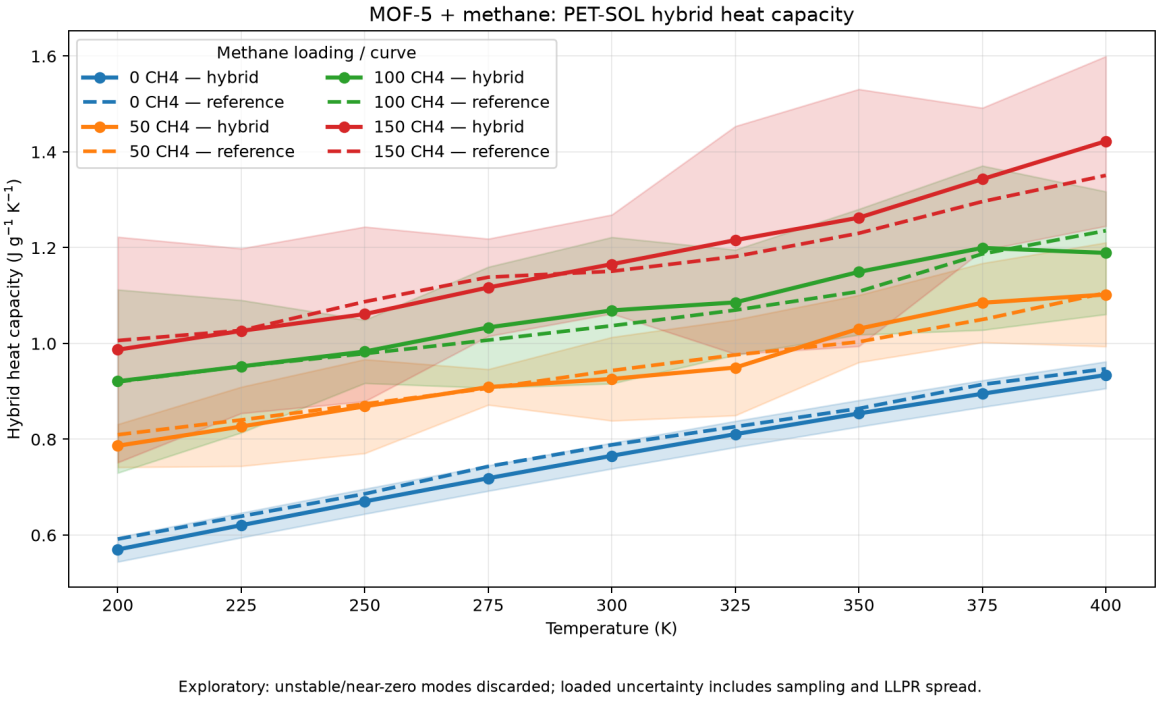


Figure 14: Supplied PET-SOL loading comparison, with the same definitions and exploratory mode treatment as Figure 13. The slight decrease of the 100-molecule central curve at the upper end of the grid is smaller than its uncertainty band and is not evidence of a resolved thermodynamic anomaly. Shading denotes standard uncertainty, not a calibrated confidence interval.

Dashed curves are digitized Kapil *et al.* data [22]. Their 300 K values are approximately 0.788, 0.944, 1.037, and 1.151 J/g/K for 0/50/100/150 guests. Loaded differences from this work are at most 0.043 J/g/K. Agreement is contextual: the digitization has no uncertainty estimates, the force fields and preparation differ, and present sampling/spectra remain unconverged. The earlier approximate empty-host scale is not a separate fitting target.

5.5 Hessian stability

Table 7: Archived central loaded-Hessian diagnostics. All calculations use PET-JAX, float64, three graph hops, and the force-constant row-sum acoustic treatment. "Converged" reproduces the relaxation archive's flag for its requested tolerance. Imaginary modes have signed frequencies below $-1\,\mathrm{cm}^{-1}$; near-zero modes satisfy $|\widetilde{\nu}| \leq 1\,\mathrm{cm}^{-1}$. Retained modes have positive frequencies above the threshold. Signed negative values represent unstable curvature, not physical negative-frequency oscillators.

Model	$N_{\mathrm{CH_4}}$	Converged	$F_{\max}$ (eV/Å)	Imaginary	Near zero	Retained	$\widetilde{\nu}_{\min}$ (cm^{-1})
PET-MAD	50	no	0.0170	17	5	2000	-53.5
PET-MAD	100	yes	0.0097	114	3	2655	-92.4
PET-MAD	150	no	0.0095	28	3	3491	-66.8
PET-SOL	50	no	0.0054	32	3	1987	-107.7
PET-SOL	100	yes	0.0100	95	3	2674	-138.4
PET-SOL	150	no	0.0049	35	3	3484	-86.5

Force convergence does not establish positive curvature: the two 100-molecule structures pass their archived force criteria but retain 114/95 imaginary modes, down to $-92.4/-138.4\,\mathrm{cm}^{-1}$. Other loaded runs have 17–35 imaginary modes and failed convergence flags; PET-MAD/50 also has five near-zero modes. Run-dependent criteria must be judged from each archive rather than comparing residual forces alone. Empty spectra have four/five imaginary modes, down to $-5.5/-6.6\,\mathrm{cm}^{-1}$.

Discarding negative eigenvalues enables an oscillator sum but does not create a stable minimum. Distinguish insufficient relaxation, unstable stationary structures, and sparse-curvature errors with convergence checks. Matching quantum/classical mode masks avoids inconsistent subtraction; it does not estimate unstable-mode bias. Reusing one 400 K geometry also leaves temperature dependence and minimum-to-minimum variation untested. LLPR spread at a central geometry does not replace structural sampling.

5.6 Uncertainty bands

Figures 15–16 separate MD sampling, classical LLPR, harmonic LLPR, and paired total uncertainty. MD errors are about 0.010 J/g/K at 300 K; model terms dominate. Covariance can make the total smaller than an individual component (Eq. (146)).

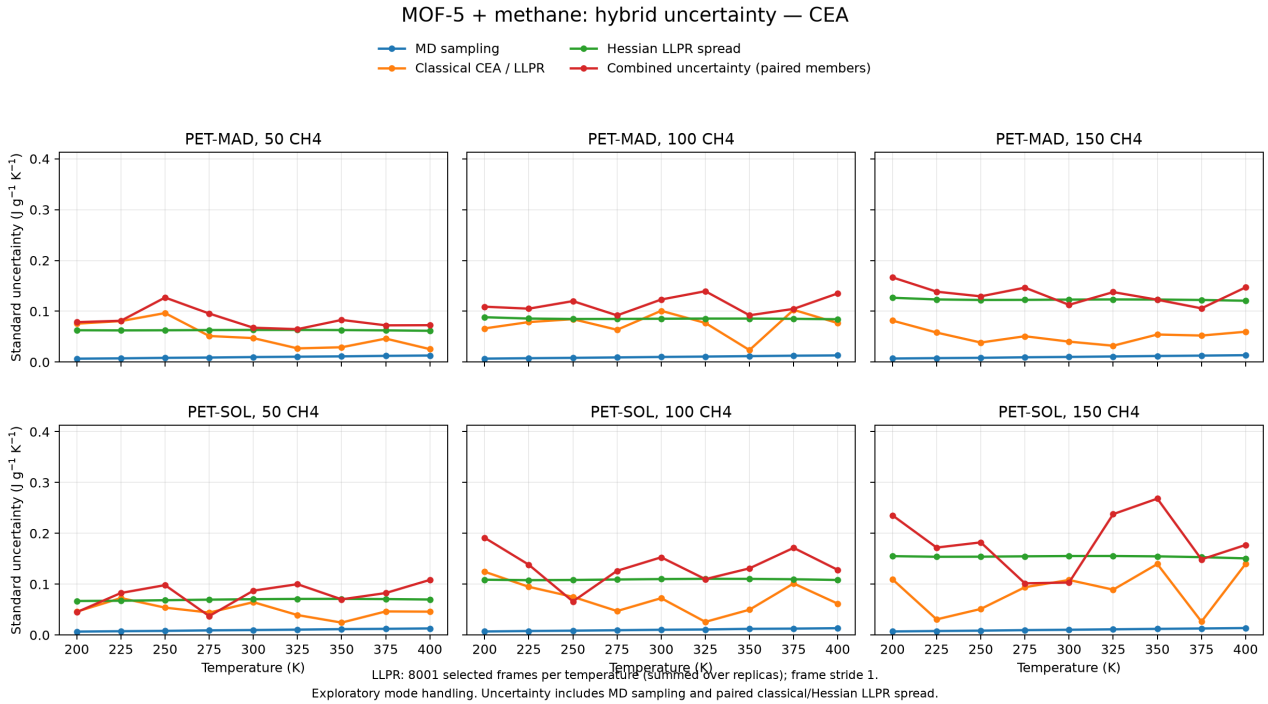


Figure 15: Supplied uncertainty decomposition using refreshed CEA member curves. Every panel uses all 8001 saved production frames at each temperature and 64 LLPR members. The harmonic curve is model spread at one fixed geometry, not an uncertainty across minima. The total preserves classical–harmonic covariance and excludes an unestimated harmonic sampling term. All plotted quantities are standard errors or model standard deviations in J/g/K, not variance fractions.

For PET-SOL with 150 guests at 300 K, classical and harmonic CEA spreads of 0.108 and 0.155 J/g/K combine to 0.103 J/g/K, with inferred correlation −0.75. PET-SOL/100 instead has correlation +0.37 and paired spread 0.153 J/g/K. Independent quadrature would miss both effects; predictive spread is not reduced by $\sqrt{64}$.

Gaussian propagation gives larger 300 K bands in every case (Table 6), with positive classical–harmonic correlations of 0.43–0.63. Smoother or narrower bands do not demonstrate calibration. Recorded $\mathrm{var}(\beta\Delta U_i)$ spans about 2.4 to 1.42×10^3, violating the small-variance CEA criterion; some direct reweighting effective counts approach one. Neither collapsed reweighting nor untested Gaussian tails validate these bands, whose coverage against reference electronic-structure data remains unestablished.

The separate `variance-hybrid` calculation also changes the central classical estimator. At 300 K, hybrid differences from the derivative route range from -0.045 to $+0.064\,\mathrm{J\,g^{-1}\,K^{-1}}$, with MD errors of 0.022–0.030 J/g/K from squared-fluctuation correlations. Shared trajectories and harmonic corrections make these cross-checks correlated. Quantitative predictions require stable Hessians, stationary independent trajectories, multiple quenched structures, and validated member propagation.

6 Recommended production and analysis protocol

Validate timestep through short NVE energy-drift tests before reservoir coupling, which can mask integration errors. Cell relaxation requires a stress-capable, validated model. One-cell Hessians and selected minima need finite-size and configurational convergence; their unweighted spread is not automatically a free-energy-weighted average. Transfer of the reference hybrid approximation to PET, especially below 200 K, needs validation.

7 Implementation, outputs, and reproducibility

`mof_heat_capacity.protocols.loaded` prepares campaigns; `scripts/md/submit_loaded_md.sh` submits loaded NPT and rejects zero-methane MD. Property stages use the scripts under `scripts/properties/`: `submit_analysis.sh` for diagnostics, `submit_heat_capacity.sh` for fixed-cell relaxation and PET-JAX Hessians, and

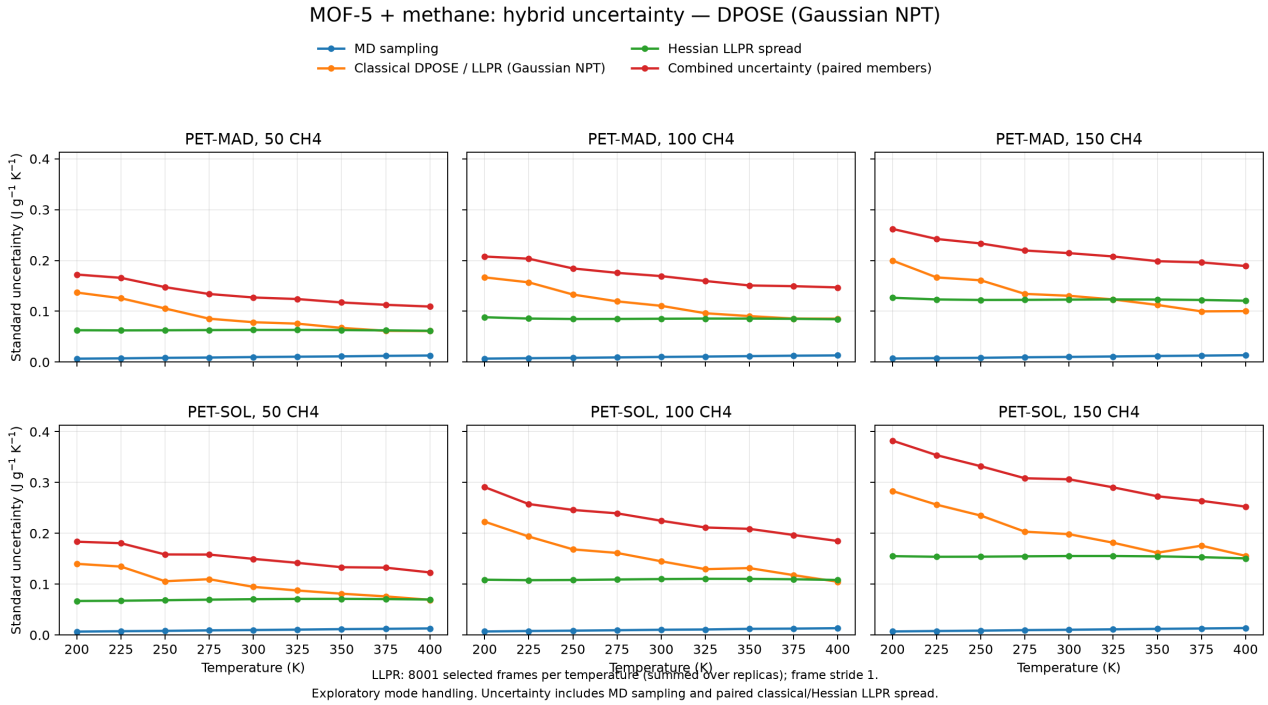


Figure 16: Supplied Gaussian NPT/DPOSE uncertainty decomposition. Central hybrid curves, MD sampling errors, harmonic member corrections, and frame coverage are identical to the CEA comparison; only the classical member spread and its covariance with the harmonic correction change. The smoother orange curves result from a different estimator and do not establish more accurate uncertainty calibration.

Table 8: Production requirements for the workflow in Figure 1. Reference agreement supplements internal convergence; it is not a fitting target.

Calculation	Acceptance and convergence requirements
Empty reference	Relax the equilibrated host directly at fixed cell with the selected checkpoint. Report forces and signed modes; retain a separately normalized harmonic curve.
Loaded NPT	Independent placement/velocity seeds. Check stationary enthalpy, cell, bonds, pores, and guest motion; converge duration, replicas, cell size, timestep, coupling times, and temperature spacing.
Loaded minima	Quench representative independent configurations at fixed cell. Reject unconverged or unstable structures.
Hessians	Converge hops, precision, HVP batching, acoustic treatment, frequency threshold, and supercell/phonon sampling on the correction.
Hybrid assembly	Average accepted loaded corrections; differentiate replica enthalpies; pair LLPR indices. Report MD, minimum sampling, and model uncertainties separately and jointly.

`submit_hybrid_analysis.sh` for final assembly. Hessian LLPR members run as eight GPU array tasks of eight members each; `analysis.llpr_merge` supplies the final hybrid dependency.

Archives retain structures, signed frequencies, model/run identities, grids, precision, hops, chunking, and rematerialization. Hybrid NPZ/CSV/JSON outputs store the classical term, harmonic correction, sum, uncertainties, and provenance.

Table 9: Uncertainty components, their calculation locations, and their stored outputs.

Quantity	Where it is calculated	Stored result
LLPR energy standard deviation	metatrain LLPR covariance and calibration	`energy_uncertainty`; copied as `analytical_energy_uncertainty_eV` in the per-run archive
CEA member enthalpies and overlap diagnostics	`analysis/uncertainty.py`, function `committee_enthalpy_estimates`	per-run `model_uncertainty.npz`
Classical LLPR C_P standard deviation	`analysis/results.py`, after member-wise replica averaging and temperature differentiation	`model_uncertainty_heat_capacity.npz` and `.csv`
MD sampling standard error	`analysis/statistics.py`, then replica combination and derivative resampling in `analysis/hybrid.py`	`classical_anharmonic_cp_standard_error`
Harmonic sampling standard error	`analysis/hybrid.py`, across independent optimized minima	`harmonic_quantum_correction_standard_error`
Hessian LLPR standard deviation	`analysis/harmonic.py`, after differentiating persistent LLPR readouts at the central minimum	`harmonic_quantum_correction_model_standard_deviation`
Combined standard uncertainty	paired model variance and sampling quadrature in `analysis/hybrid.py`	`approximate_cp_combined_standard_uncertainty`

The scalar energy uncertainty is diagnostic; property errors require complete persistent member curves. Paired MD/harmonic readouts preserve covariance, but Hessians use central-model minima without member-specific relaxation.

Use one node, one task, and one GPU per MD/Hessian job; parallelize independent states, replicas, minima, or member tasks. Retain Slurm IDs and measure representative runtime/memory before adjusting requests:

```
sacct -j JOB_ID --format=JobID,State,Elapsed,MaxRSS,AllocTRES
```

Use distinct outputs for every loading, temperature, seed, and replica.

Record structure/composition and insertion criteria; checkpoint, export, conversion, and environment identities; LLPR dataset/ensemble hashes, calibration, regularizer, seeds, member count, stride, and overlap diagnostics; ensemble/integration settings and restart history; relaxation forces, signed modes, and Hessian controls; and production intervals, replica counts, correlation/block methods, and convergence comparisons.

8 Limitations and interpretation

Table 10: Limits of the present estimates and the evidence needed to resolve them.

Present limitation	Required evidence
33/66 enthalpy drift flags; one replica per state	Longer independent trajectories and stable block/replica statistics.
Imaginary modes in every loaded reference; four failed relaxations	Stable minima and converged signed spectra, sparsity reach, and precision.
One loaded geometry per curve	Multiple independent quenches and sensitivity to harmonic-reference temperature.
Classical MD plus harmonic C_V correction	Validation of quantum anharmonic and pressure–volume effects, especially below 200 K.
Fixed-loading random insertion	Demonstrated equilibration of guest distributions; separate particle-exchange sampling for adsorption equilibrium.
Shallow LLPR; large CEA diagnostics and method-dependent bands	Reference-data calibration and validated propagation; shared representation bias and member-specific relaxation remain outside the current spread.
PET differs from QuickFF; finite-cell/numerical approximations	Loaded-MOF energy/force/stress/curvature benchmarks and size/grid convergence.

9 Conclusions

PET-MAD and PET-SOL give exploratory room-temperature heat capacities rising from about 0.93 to 1.16 J/g/K with methane loading. Harmonic quantization removes 59–63% of the classical term; guest mobility supports retaining anharmonic MD. Broad agreement with digitized reference trends does not establish quantitative accuracy. Stable spectra, independent trajectory/minimum sampling, and validated paired uncertainty propagation are needed before interpreting these curves as converged predictions.

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